
RECURSIVE HORIZONS

A Thermodynamic Research Programme for
Black-Hole-to-Cosmology Transitions

Can Einstein and Hawking Finally Rest Easy Together? Here Is What Must Be Proven

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BLACKHOLES-INFINITY PROJECT
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Contents

Abstract	5
Plain-language summary	8
1. The epistemic contract	8
2. Why gradients are a productive starting point	9
2.1 Difference makes dynamics observable	9
2.2 From gradient language to a model	9
2.3 Finite Gradient Closure as a fail-capable law	10
2.4 The analogies—and their limits	11
2.5 DSF-1: constants as domain settings, stated operationally	11
2.6 We see things for what they truly are: open-minded without madness	12
3. Causal speed, photons, and the meaning of space	13
3.1 c is not produced by light sources	13
3.2 Infinite extent does not require infinite speed	13
3.3 Finite-domain causal calibration and the VSL branch	14
3.4 What classical black-hole geometry establishes	15
3.5 The Planck-scale localization argument	16
4. What general relativity actually contributes	16
4.1 A black-hole interior is cosmological in a precise sense	16
4.2 The Hubble-sphere compactness identity	16
4.3 Why a black hole does not consume every available energy density	17
5. Horizon thermodynamics and the conditional relation	17
5.1 Established area laws	17
5.2 The new saturation postulate	18
5.3 Conditional derivation	18
5.4 What the numerical example means	18
5.5 Rotation	19
6. Candidate transition mechanisms	19

6.1 Einstein–Cartan gravity	19
6.2 Loop quantum cosmology	19
6.3 Tunnelling and baby universes	20
6.4 The transition is the theory	20
6.5 CCT-1: a controlled covariant core	20
6.6 DST-1: one existing scalar, three established regimes	26
7. The information problem	27
7.1 A map, not a slogan	27
7.2 IM-1: an explicit code-subspace isometry	27
7.3 Inheritance of constants	28
8. Observational reality in 2026	28
8.1 The standard cosmological baseline	28
8.2 DESI DR2	29
8.3 Planck isotropy and directional spectral index	29
8.4 Local causal-speed constraints	29
8.5 Domain-setting and dark-sector tests	30
9. What the existing computations do—and do not—show	32
10. The minimal Recursive Horizons hypothesis	33
RH-0: Finite Gradient Closure	34
RH-1: Effective causal phase	34
RH-2: FGC restoring transition	34
RH-3: Information-preserving map	34
RH-4: Effective child cosmology	34
RH-5: Optional horizon saturation	34
11. How to prove the proposal wrong	35
11.1 Mathematical failure	35
11.2 Cosmological failure	35
11.3 Predictive failure	35

12. A concrete calculation programme	36
12.1 Background	36
12.2 Perturbations	36
12.3 Information	37
12.4 Observation	37
13. The philosophical layer	38
14. Why Einstein and Hawking could rest easy together	39
15. Conclusion	39
References	46

Status statement. *This is a conjectural research programme and position paper, not a proof that our universe is inside a black hole, not an established theory of quantum gravity, and not a report of an independently detected signal. NGS-0 executes only finite-band, recurrence-eigenvalue, and finite independent-trial probability controls; it derives no nested cosmology, dark sector, inevitable observer, or plan. DSF-3 defines a finite-domain causal-calibration hypothesis but supplies no new numerical result: no artifact derives a changing locally measured c , a cross-domain cone ratio, an infinite-speed layer, a CCC crossover, or superluminal entanglement signalling. Version 0.11 fixes the minimal-ECD signature/action bridge and local interaction projection, then verifies the algebraic spin/torsion/contact sector. A corrective tetrad audit retracts KIN1's earlier fixed-coordinate-current NEC violation: the axial-current response cancels the apparent anisotropy, and the physical contact stress is metric proportional with zero null contraction. One positive-frequency free-Dirac plane wave is retained only as a bounded control. The healthy restoring action, complete torsion-free Dirac source, global harmonic closure, regular-centre constraints, finite-mass exterior, curvature evolution, null defocusing, bounce, horizon, and global parent-child spacetime remain open. Neither these gates nor the earlier GMF-1, EC-1, BAO-only, or GW timing results establishes an external origin for dark energy, an S^3 child, a realistic primordial spectrum, or observational confirmation.*

Live development routing. *This versioned prospectus does not mirror the repository's append-only implementation history. The current checkpoint, active numerical frontier, compact verification commands, and closed operations are maintained in the [active code map](../docs/active-code-map.md). Exact proposition status is in the [claim ledger](../docs/claim-ledger.md), and the forward dependency path is in the [research roadmap](../docs/research-roadmap.md). Later numerical certificates do not rewrite the scientific scope of this paper: no current artifact establishes a global transition, singularity resolution, retained- EFT evolution, a child universe, a dark-sector replacement, or observational confirmation.*

+## Authorship and AI contribution disclosure

This manuscript and its executable research artifacts were developed through an iterative collaboration between **Douglas Ek and ChatGPT 5.6 Sol**, working through OpenAI Codex. Douglas Ek supplied the research direction, organizing hypotheses, conceptual synthesis, and continuing scientific decisions; the AI collaborator translated those aims into formal language, code, derivations, tests, literature comparisons, and adversarial audits. Every AI contribution was treated as candidate work to be checked against explicit mathematics, reproducible computation, cited sources, and stated nonclaims. Douglas Ek retains human responsibility for publication and for the scientific claims. The AI credit records material intellectual and production assistance; it does not imply that the system can independently consent to authorship or assume scientific accountability.

Abstract

We investigate whether a collapsing black-hole interior can pass through a finite transition into an expanding cosmological domain while retaining the established local content of general relativity, quantum field theory, thermodynamics, and Λ CDM. The central synthesis is that a causal domain may be a stable phase: its geometry, vacuum state, matter excitations, transport, and information budget may be different limits of one covariant system. Collapse and expansion could then be opposite branches of one process rather than rival descriptions. This proposition joins established ingredients in a new global relation; it is not already implied by them.

We call the guiding thermodynamic conjecture **Finite Gradient Closure (FGC)**. It distinguishes a completed infinite state from unbounded continuation. A singular endpoint and an actually completed infinite physical domain are treated as explanatory failures; an unending sequence of finite nested gradients need not be. “Multiverse” here therefore means gradients within gradients: every realized domain and transition must carry finite state data, possess a lower closure and an upper closure or finite handoff, and satisfy local conservation and the appropriate entropy law, even if the generative sequence has no final step. This is a selection principle to be implemented and tested, not a proof that mathematical infinity or every standard asymptotic construction is unphysical, and not a license for perpetual work extraction from equilibrium.

A divergent invariant, geodesic incompleteness, or failed extension may mark the boundary of a retained description, but it does not encode what lies past that boundary. An infinite decimal expansion likewise represents one number; its repeated digits are not a catalogue of successor spacetimes. Evidence for recursive continuation requires additional covariant dynamics that maps finite predecessor state through a finite transition to finite successor state while closing constraints, flux, entropy, and causal orientation. The map must then be shown composable. A singular GR control is therefore a baseline, not a positive transition result.

The motivating intuition is gradient based: observable states require differences, fluxes respond to gradients, and stable regimes can arise from competing terms. DSF-1 translates “constants as domain settings” into an operational hypothesis about dimensionless couplings, mass ratios, thermal ratios, and relative characteristic cones. It does not assert that mathematical constants vary, that a unit-dependent value of c is observable across domains, or that local light speed has already been shown to change. A conformal metric change leaves the null cone invariant; a physical speed claim requires a nonconformal characteristic structure and a dimensionless comparison such as $c_{\text{mode}}/c_{\text{gamma}}$.

The subordinate Nested Gradient Spectrum hypothesis uses literal QFT and statistical-physics language: modes, spectral density, bandwidth, occupation, resolution, and coarse-graining. It asks whether finite localization opens a richer but still finite accessible spectrum while coarser enclosing levels impose boundary and capacity constraints. Unresolved levels may be tested as effective dark sources, but no dark-sector identification follows from the idea. A universal “one hertz per black hole,” perfect equilibrium, or golden ratio is not assumed; each requires an observer-independent spectral or recurrence derivation.

The methodological posture is **we see things for what they truly are: open-minded without madness**. Dark matter and dark energy retain their observational burdens without being promoted into uniquely established microscopic substances. A gradient or relational replacement receives the same scrutiny: every explanatory variable must have an operational map, alter a predeclared prediction, recover the Λ CDM-plus-GR evidence where it succeeds, and expose a falsifier. Directly observed antimatter is not a dark placeholder, and stable asymmetry is a dynamical stability/throughput target rather than an axiom that guarantees complexity.

NGS-0 makes the presently available mathematics executable. It verifies the finite stationary scale of the spatial $-\alpha \quad k^2 + \beta \quad k^4$ control, reports the positive eigenvalue of $L_{(i+1)} = aL_i + bL_{(i-1)}$ for golden and non-golden fixtures, and evaluates $1 - (1-p)^N$ for one declared finite independent-trial ensemble. These are D-C controls, not a nested cosmology or probability model for observers.

DSF-3 now turns the associated variable-speed intuition into a separate H/Q causal-calibration problem. A lapse or scale factor changes a coordinate light slope but not the local common-cone ratio measured with proper distance and proper time. Physical variation requires action-derived principal symbols and a dimensionless relative cone; comparing successor domains additionally requires a transition map for observers, clocks, phases, sectors, and states. The admissible “top-level” target is therefore a large but finite mapped relative cone, not a completed infinite speed. CCC, background-dependent QED photon propagation, and VSL models are neighboring mathematical resources; none makes density, complexity, conformal crossover, or entanglement evidence for that target without the missing dynamics [65--71].

Version 0.6 makes that distinction data facing. A Bekenstein-type disformal electromagnetic metric provides a covariant example in which a homogeneous order parameter changes the photon cone relative to the gravitational/tensor cone [43]. DSF-2 computes the invariant ratio $\Delta T = c_T/c_{\text{gamma}} - 1$ and retains the common-cone stationary limit. Four exact-hash GWOSC/Fermi/GCN inputs reconstruct the GW170817/GRB 170817A geocentric delay as 1.737774 s, consistent with $+1.74 \pm 0.05$ s; the quoted relative-speed interval remains conditional on the collaboration's source-emission-lag bracket [26].

DEB-1 separately maps an explicit boundary density scaling to its effective equation of state. The public 13-observable DESI DR2 BAO vector and full covariance are now evaluated in a declared flat late-time constant- w Gaussian profile [25]. Its minimum is $w = -0.911899$, $\Omega_m = 0.297693$, and $\chi^2 = 9.0410$. A constant density ($w = -1$) is worse by $\Delta \chi^2 = 1.2300$, a constant-tension surface-energy ansatz ($w = -2/3$) by 11.3325 , and an inverse-area density ($w = -1/3$) by 69.8429 ; the last is also nonaccelerating in GR. These results reject two literal conserved background scalings in the stated model. They contain no information about whether the surviving near-constant component is internal or external.

DST-1 then reuses the canonical scalar already present in the controlled model. For $V=V_0+m^2 \varphi^2/2$, a nonzero stationary offset is exactly vacuum-like, $w=-1$, while rapid quadratic oscillations have the established averaged limit $w=0$ and ρ_{osc} proportional to a^{-3} [41]. A matched source pair $Q=\Gamma \cdot \varphi^2$ conserves total background energy while transferring scalar energy to radiation, but Γ remains phenomenological until interactions and thermalization are derived. These equations demonstrate that one field can contain vacuum-like, matter-like, and transfer regimes. They do not derive the observed dark-energy scale, dark-matter clustering, or the hot Big Bang.

General relativity also supplies a precise example of competing geometric tendencies. In the weak-field Schwarzschild–de Sitter/Kottler limit, $a_r = -GM/r^2 + \Lambda c^2 r/3$, with a balance radius $[3GM/(\Lambda c^2)]^{1/3}$ [44]. This shows that compact-mass attraction and negative-pressure vacuum acceleration can coexist and define an unstable sign-change scale. It is not a stable equilibrium, an accretion wall, or a parent-domain proof: black-hole growth is governed by actual matter and gravitational flux across a horizon [45]. The stronger claim—that additional covariant dynamics can create and stabilize recursive causal domains—remains the global calculation to be done.

As a controlled realization, we study Einstein gravity with a canonical scalar at a positive-potential stationary point on closed FLRW slices. Its vacuum exact solution, $a(t)=L \cosh(t/L)$, is global de Sitter space in closed slicing: curvature invariants are finite, the scalar and tensor kinetic terms are healthy, and the two radial null expansions derive normal, trapped, marginal, and anti-trapped spheres. This is a covariant baseline, not the desired black-hole bridge. GMF-1 now makes one obstruction exact: at the same Λ , the finite Schwarzschild/Kottler mass is a Misner–Sharp jump relative to the pure core, so no direct smooth Darmois attachment exists [48,50]. Its constant invariants expose an important limitation: the scale-factor bounce alone is a property of the selected cosmological congruence, not evidence for new transition microphysics.

GMF-1B-ECD-SYM1 then fixes the smallest viable axial-current preflight after two shortcut closures fail. For the published Ventrella–Choctui massless left-chiral spherical pair [62], the total temporal and radial axial current survives the two-field sum and its unit-normalization invariant is $A_a A^a = -16|F|^2|G|^2$; the declared equal-profile full-Dirac singlet comparator instead cancels that axial channel [63]. INT1 makes the convention bridge explicit. Starting from the canonical (+---) torsion-eliminated interaction $-3 \kappa J^2/16$ [64], $\Gamma^a = -i \gamma_{VC}^a$ and $J^a = -i \bar{\psi} \gamma_{VC}^a \gamma_{\text{star}} \psi$ give the translated VC (-+++) interaction $+3 \kappa A^2/16$ and the corresponding Hehl–Datta equation [61,64]. The executable verifies that variation of the sphere-reduced contact action agrees with the directly projected cubic coefficient and that the local four-component output reconstructs in the declared two-vector basis at three tested angles. TOR1 verifies the spin tensor and algebraic torsion invariant. Corrected KIN1 then checks all 16 coframe variations: the coordinate-current response cancels the false fixed-current anisotropy, leaving a metric-proportional contact stress with zero null contraction. The earlier local NEC claim is retracted. Because the interaction contains no derivatives, it leaves the Dirac principal cone metric-null. Either one-amplitude control retains a nonzero null axial current and algebraic torsion source even though the contact invariant and projected cubic source vanish. A normalized smooth annular packet supplies exact initial spinor-zero buffers. These are local axial-bilinear, convention, contact-stress, and initial-profile facts, not full-source equivariance, the still-missing torsion-free Dirac stress, constraint solution, defocusing history, bounce, or global spacetime.

We isolate an optional entropy-saturation postulate: the accessible asymptotic de Sitter entropy of a child equals the Bekenstein–Hawking entropy assigned to its parent horizon. For spherical horizons this gives $\Lambda = 3/r_s^2$. The algebra is exact, but its empirical status is limited: when the observed Λ determines r_s , the resulting parent mass is an inference and resubstitution is an identity, not a prediction. We correct earlier claims concerning variable local light speed, photons before stars, infinite space, minimum length, universal torsion bounces, CMB anisotropy, and particle filtering. Existing repository CMB and Kantowski–Sachs computations are classified as invalid or unfinished.

The first information artifact is an exact isometry on a stated finite semiclassical code subspace. It demonstrates nonduplicating fine-grained transfer but does not factorize the gravitational Hilbert space or imply equality of horizon entropies. Linear scalar-field and tensor modes are propagated through a finite conformal-time core with a symplectic transfer matrix; no n_s , A_s , or particle-production claim is made because an initial state, reheating history, and adiabatic curvature mode have not been derived. An additional globally closed-child branch postulate supplies one independent conditional sign discriminator, $\Omega_K < 0$; CCT-1 alone does not connect the closed core slicing to our observable global topology. A robust future $\Omega_K > 0$ inference that survives stated nonflat-model and systematic

checks would reject that branch; a result statistically consistent with zero would not confirm it because later expansion can suppress its magnitude.

The remaining scientific burden is explicit: derive the domain-setting functions and relative cones; embed the core in a horizon-regular black-hole spacetime with a real flux/junction balance; replace the stationary baseline by controlled transition dynamics; derive abundance, reheating, the adiabatic perturbation spectrum, and structure growth; and predict a dimensionless magnitude or correlation fixed independently of the data used to test it. The proposal is worth developing precisely because it can be made to fail without asking established physics to fail with it.

Plain-language summary

The image behind this project is simple. A black hole is not merely a dark ball in space; it is a causal region whose interior geometry evolves differently from its exterior. General relativity describes that interior as a collapsing anisotropic spacetime. Quantum-gravity models sometimes replace singular collapse with a bounce. Horizon thermodynamics assigns finite entropy to black-hole and cosmological horizons. Those facts invite a question: could collapse, a finite transition, and expansion be consecutive phases of one causal process?

Version 0.7 sharpens the proposed answer by restricting what “one globally connected solution” must mean. A “domain” is not a sphere drawn inside another sphere. It is a regime in which geometry, causal cones, vacuum stress, field excitations, and transport settle into a reproducible relationship. A local core, a parent metric, and a drawn shell do not become one solution until the matching, quasi-local flux, causal structure, and stability equations are solved. The attraction of this picture is not that it discards Einstein, Hawking, thermodynamics, quantum fields, or Λ CDM. It asks whether their successful equations are the local limits of one larger process.

That idea already has calculable pieces. A positive scalar-potential offset behaves like vacuum energy; rapid quadratic scalar oscillations behave like pressureless matter on average; a balanced source term can transfer their energy into radiation; and positive- Λ geometry can oppose compact-mass attraction at a definite radius. The repository now computes these facts. Their union is a research direction, not yet an identification of the cosmic dark sector or a proof of ancestry.

That question is reasonable. Its answer is not already contained in the ingredients. A coordinate change showing that the Schwarzschild interior resembles a cosmology does not create a child universe. A bounce in a symmetric model does not establish a realistic black-hole bounce. Unitarity does not equate two coarse-grained horizon entropies. A numerical equality produced by solving for an unknown parent mass from the observed cosmological constant is not a prediction of that constant.

The useful move is to keep the question and place certainty exactly where the equations earn it. We can then ask what field selects the effective geometry and dimensionless settings, what balance stabilizes a domain, what changes that balance, how null rays pass from trapped to expanding regions, how the hot plasma and clustered matter arise, how information crosses the transition, and what observation would differ from standard cosmology. That is a grand proposal in a form a physicist can calculate, criticize, or falsify.

1. The epistemic contract

The purpose of this work is to match reality, not to win an argument. We therefore use six categories throughout:

Label	Meaning
Established	A result of a mature theory or observation, restricted to its demonstrated domain
Observed	A published constraint with a named dataset, release, model, estimator, and uncertainty
Derived	An identity or conditional consequence of explicitly stated premises
Hypothesized	A new physical postulate introduced here
Analogy	A reasoning aid that does not itself supply gravitational dynamics
Open / negative / retired	A missing derivation, failed test, invalid method, or abandoned branch

Three distinctions matter especially.

First, an **identity** is not an independent prediction. If x is defined from observed y using $y=f(x)$, then computing $f(x)$ returns y by construction.

Second, a **coordinate statement** is not necessarily a local measurement. Schwarzschild-coordinate dr/dt can approach zero at a horizon while a freely falling observer measures nearby light at the invariant speed c .

Third, a result established in a **symmetry-reduced model** is not automatically generic. A homogeneous isotropic LQC bounce or a spin-fluid ECKS bounce demonstrates possibility under stated assumptions; it does not prove the outcome of rotating, inhomogeneous stellar collapse.

This contract does not make the project timid. It makes the ambitious part visible.

2. Why gradients are a productive starting point

2.1 Difference makes dynamics observable

The project's central intuition is that a perfectly featureless state cannot define observable change. Physics supports a precise, limited version of that claim.

In quantum mechanics the Hamiltonian generates evolution [27],

$$i \hbar \partial_t |\psi\rangle = H |\psi\rangle,$$

and observable phase evolution between stationary states depends on an energy difference,

$$\omega_{nm} = (E_n - E_m) / \hbar.$$

An overall shift of all energies is normally unobservable in nongravitational quantum mechanics; differences generate relative phase. In linear nonequilibrium thermodynamics, gradients of thermodynamic potentials act as generalized forces that drive corresponding fluxes [29]. In field theory, gradients contribute kinetic energy and propagation.

The scope matters. “Energy generates time” is not a universal global theorem in arbitrary curved spacetime. A conserved energy arises from time-translation symmetry [28]; generic expanding FLRW spacetime has no global timelike Killing vector and no unique globally conserved total energy. Local stress-energy conservation, $\nabla_\mu T^{\mu\nu} = 0$, remains the appropriate statement.

The programme can therefore begin with a field or order parameter Φ , its gradients, and a dynamical law. It cannot begin and end with the word “gradient.”

2.2 From gradient language to a model

A minimal effective description would take a form such as

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{2\kappa} + \mathcal{L}_\Phi(\Phi, \nabla \Phi) + \mathcal{L}_\text{matter} + \mathcal{L}_\text{interaction} \right].$$

If geometry is emergent rather than fundamental, the model must instead define an effective metric,

$$g_\text{eff}(\mu, \nu) = G_{\mu, \nu}[\Phi, \nabla \Phi, \text{state}, \dots],$$

and demonstrate that perturbations obey hyperbolic equations with a stable Lorentzian signature. Analogue-gravity systems show that perturbations in a medium can experience effective horizons and metrics [16,17]. Jacobson showed that Einstein's equation can be read as an equation of state under specific local horizon assumptions [4]. These are legitimate precedents for asking whether geometry is collective. They do not establish this particular substrate.

2.3 Finite Gradient Closure as a fail-capable law

FGC makes the gradient intuition conditional and testable. When gravitational localization steepens past a dynamically derived invariant threshold, a healthy higher-order geometric or matter response must acquire the restoring sign in the physical collapse channel and grow faster than the lower-order focusing term. It must dominate at a nonzero finite scale before geodesic incompleteness or effective-theory breakdown, while preserving constraints, causal propagation, flux, and generalized entropy. No matter sector or black-hole genealogy is selected by this statement alone.

Two toy controls expose the mathematical burden. A spatial mode energy

$$E_k \text{ proportional to } (-\alpha k^2 + \beta k^4) |\Phi_k|^2, \\ \alpha > 0, \beta > 0,$$

selects $k_*^2 = \alpha/(2\beta)$ and $\ell_* = \sqrt{2\beta/\alpha}$. An effective radius potential

$$V_{\text{eff}}(R) = -A/R + B/R^n + C R^m, \\ A, B, C > 0, n > 1, m > 0,$$

can bound finite-energy motion from both below and above. These controls prove only that competing orders can select finite scales. A relativistic implementation must independently avoid higher-derivative ghosts—preferably through a second-order auxiliary regulator unless degeneracy is proved—and derive rather than insert its terms.

The local causal gate is sharper. For an affinely parametrized, irrotational null congruence,

$$d\theta/d\lambda = -\theta^2/2 - \sigma_{ab}\sigma^{ab} - R_{ab}k^a k^b.$$

A trapped-to-expanding transition requires the restoring sector to overcome the expansion and shear terms for a finite interval, or to derive why a different microscopic causal equation applies. A coordinate scale-factor minimum is not that proof. Nor is lower closure sufficient: the same solution must provide an upper closure, asymptotic operational completion, or finite handoff with finite quasi-local state data. An unbounded genealogy can then be an unending **process** of finite transformations without becoming a completed infinite state or a perpetual-motion engine.

Nested Gradient Spectrum and scale flow

The spectral language is not merely metaphorical. QFT expands fields in modes of a declared operator/background, spectral functions encode response, and effective field theory plus renormalization organize descriptions by resolution. Thermodynamics supplies state occupation, free energy, transport, and entropy. NGS asks whether an FGC transition defines an actual mode-transfer map

$$g_{(i+1)}(\omega) = T_{i\{g_i\}}(\omega) + \Delta g_i(\omega),$$

with a finite change in accessible mode count and a dimensionless frequency or spacing prediction. Proper frequency must be defined relative to an observer, so a universal SI-hertz increment is not invariant.

Competing gradient, higher-gradient, nonlinear, boundary, and possibly dissipative terms can select a finite band of structure. Dissipation is not a universal prerequisite: conservative and unitary systems also generate complex correlations, while friction can either stabilize or erase a pattern. The general test is whether the equations derive the instability, saturation, restoring response, and entropy production claimed.

The golden ratio is retained only as a candidate eigenvalue. If the derived cross-level scales obey $L_{(i+1)} = L_i + L_{(i-1)}$, their asymptotic ratio satisfies $q^2 = q + 1$ and hence $q = (1 + \sqrt{5})/2$. Friction by itself does not produce that recurrence. The transfer dynamics must derive it before the ratio is compared with a spectrum or population.

NGS-0 records these three mathematical controls with explicit nonclaims. It does not derive the recurrence coefficients, domain ensemble, cross-level mode operator, or any physical dark component.

An additional NGS hypothesis asks whether a degree of freedom observed in a daughter domain as a particle could be a pole, resonance, or spectral band induced by coupling to modes of an inherited or adjacent phase. That is a possible

change of ontology, not a denial of the measured phenomena. Quark and Higgs signatures, masses, decays, scattering amplitudes, confinement behavior, and detector responses remain compulsory evidence. A deeper account would have to derive a transition kernel and daughter spectral density whose poles, widths, residues, charges, spin/statistics, unitarity, and Lorentz behavior recover those data while also freezing a new discriminator. Without that operator and prediction, “a remnant from a previous gradient” remains an analogy.

The introductory billiard-ball picture is therefore not the target being defended: QFT already describes particle content through quantized field excitations, states, correlators, amplitudes, and localized detector events. NGS adds the unproved possibility that a common finite-gradient substrate derives localization and persistence. It cannot assume that every excitation is a soliton; that would require a nonlinear finite-energy solution, an actual stability mechanism, and a quantization map recovering observed spin and statistics. Nor does ordinary nuclear binding-energy release by itself prove a deeper pressure medium. “Pressure” must become a derived stress tensor or energy functional before it carries explanatory weight.

This hypothesis is deliberately outside the present SF1 calculation. SF1 uses an independent classical scalar χ as a controlled collapsing stress carrier and ϕ as the candidate regulator. It contains no neutron-star equation of state, quark/Higgs microphysics, particle-destruction rule, or cross-domain spectral-transfer operator.

2.4 The analogies—and their limits

Several analogies in the historical corpus are genuinely generative:

- **A trampoline or thin membrane** suggests an order parameter whose stiffness and curvature vary with state.
- **Cavitation and pressure chambers** suggest phase instability, nucleation, and a boundary separating regimes.
- **Acoustic horizons** demonstrate that a finite characteristic speed and effective causal cones can emerge for perturbations in a medium [16,17].
- **Quantum tunnelling** shows that a classically forbidden connection can have nonzero transition amplitude; baby-universe tunnelling has a theoretical history [15].
- **Parametric resonance and energy pumping** suggest how periodic background structure can transfer energy into selected modes.
- **Microfluidic sorting** demonstrates that gradients and interfaces can separate populations by response coefficients.
- **Biological repair and self-correcting codes** suggest feedback, redundancy, and error thresholds.

None of these is a derivation of black-hole cosmology. A fluid has constitutive equations; a membrane has material degrees of freedom; a tunnelling calculation has an action and boundary conditions; a sorter has a known force law. To graduate from analogy, each picture must identify the corresponding invariant variables, equations, and conserved quantities in the gravitational model.

This particularly affects the retired “torsion filter.” Microfluidic selection by spin is an analogy until an action produces a spin-dependent transition amplitude and a cosmological abundance. Minimal Einstein–Cartan theory does not by itself delete scalar superpartners or import neutralinos.

2.5 DSF-1: constants as domain settings, stated operationally

The stronger thesis is not that reality contains no stable quantities. It is that some quantities which appear constant inside one causal domain may be vacuum or state values selected by dynamics. Bekenstein’s covariant varying- α programme and his general relation between physical and gravitational metrics show that questions of this type can be formulated without abandoning covariance or causality [42,43]. They do not establish that any such variation occurs in Recursive Horizons.

The first discipline is categorical. Mathematical constants such as π do not vary as material couplings. The numerical values of dimensional quantities such as c , G , $\hbar$, a temperature in kelvin, or the Planck length depend on units. A cross-domain claim becomes physical only through an operational dimensionless comparison, for example

```
X_domain = { alpha,
              m_e/m_p,
              G m_p^2/(hbar c),
              k_B T/(m_e c^2),
              c_T/c_gamma,
              entropy and transition ratios }.
```

The Planck length $\ell_P = \sqrt{\hbar G/c^3}$ is a derived dimensional combination, not an observed lattice spacing. Absolute zero is a thermodynamic ground-state limit, not zero quantum energy or the disappearance of geometry. Effective couplings can also run with renormalization scale without varying in cosmic time. DSF-1 keeps these objects separate.

The second discipline is dynamical. For a mode a , an action must derive a characteristic equation

```
G_a^{(\mu \nu)}(\Phi, \nabla \Phi, ...) k_\mu k_\nu = 0.
```

If $G_a^{(\mu \nu)} = \Omega(\Phi)^2 g^{(\mu \nu)}$, the null cone is conformally unchanged; no physical speed variation follows. A nonconformal characteristic structure can instead define an observable such as

```
delta_a(z, k, n) = c_a/c_gamma - 1.
```

The model must derive its sign, scale, direction and frequency dependence, recover the common observed cone in the accessible domain, and freeze its parameters before the target data are examined. DSF-1 is therefore an operational specification of RH-1, not an extra proof of the transition.

2.6 We see things for what they truly are: open-minded without madness

The project is intended to escape paradigm lock without escaping evidence. That requires a symmetric rule: neither an inherited noun nor a preferred new picture is allowed to do explanatory work merely because it has been named. “Dark matter” and “dark energy” compress linked observational behavior inside specified models; they do not, by themselves, prove a unique particle species, literal vacuum substance, modified geometry, or extra-domain origin. But the phenomena are not optional. CMB structure and lensing [19], the separation of lensing mass from most baryonic mass in the Bullet Cluster [72], and the two independent supernova programmes that established late acceleration [73,74] are among the constraints a replacement must jointly recover. Λ CDM plus GR is therefore the empirical comparator wherever it works—not a protected ontology, and not a baseline this programme may ignore.

The anti-epicycle constraint is **no unaccountable explanatory variable**, not “no latent variable.” Physics routinely infers fields and parameters before direct particle detection. For every proposed degree of freedom q , the model must instead publish a variable ledger,

```
q = {observable/latent/nuisance/gauge/interpretive status,
     units and invariant outputs,
     measurement or transition map,
     symmetry redundancy and prior domain,
     calibration data, held-out target, falsifier},

y = H(q, theta) + measurement error.
```

An added physical branch must change at least one predeclared operational distribution,

```
p(D_*|M_1) != p(D_*|M_0),
```

and the target D_* cannot be the same information used to invent or tune the branch. A named boundary force, cross-domain pull, hidden gradient, or freely chosen function that never changes a held-out prediction is not an explanation. Nor is a model simpler merely because it replaces two nouns with one unconstrained relation. A unified dark-sector claim must reproduce both the matter-side CMB/BBN, clustering, lensing, growth, structure, and halo burdens and the acceleration-side supernova, BAO, CMB-distance, growth, local-gravity, stability, and propagation burdens with one action and parameter ledger.

Antimatter marks an important evidentiary boundary. The positron was observed directly [75]; antihydrogen is produced, trapped, and spectroscopically tested [76], and its free fall has been observed to be consistent with ordinary terrestrial attraction at present precision [77]. The open cosmological problem is the matter–antimatter abundance asymmetry, together with sharper CPT and equivalence tests—not whether antimatter is an unresolved dark noun.

The word “relational” is used in an interpretation-neutral operational sense. Claims should be expressed through gauge/diffeomorphism-invariant quantities or explicit relations among clocks, rods, detectors, phases, sectors, and transition maps. This is compatible with Rovelli’s Relational Quantum Mechanics [78], but does not assume its interpretation of quantum states. A quantum interpretation cannot supply a missing gravitational action or shield FGC from a failed prediction.

Finally, the proposed creative engine is sharpened from “imbalance must create complexity” to **stable asymmetry**. For a declared state x , order parameter $A(x)$, and dynamics

$$\dot{x} = F(x; \lambda, J),$$

an asymmetric fixed attractor requires

$$\begin{aligned} A(x_*) &\neq 0, \\ F(x_*; \lambda, J) &= 0, \\ \max \operatorname{Re} \operatorname{eigenvalue}[D_x F(x_*)] &< 0. \end{aligned}$$

A periodic attractor requires the corresponding transverse Floquet test. If the claim is specifically a driven nonequilibrium engine, its declared fluxes, affinities, reservoirs, work terms, and coarse-grained entropy must also give nonnegative finite entropy production, with at least one active throughput channel, on every finite operational interval. This does not hardcode mechanical friction. Symmetric laws can support asymmetric phases, and an equilibrium symmetry-broken phase can be structured with zero sustained throughput. Life-like persistence still consumes free energy and exports entropy, but dissipation alone does not prove life’s origin. The equations must decide which version, if any, reality realizes.

3. Causal speed, photons, and the meaning of space

3.1 c is not produced by light sources

In relativity, c is the local invariant speed defining the null cone of a Lorentzian metric. A photon follows a null path, locally $ds^2=0$, but the causal structure does not disappear when no photon is present. Stars are sources of stellar photons; they did not create the constant c .

The early universe was not dark because photons did not exist. It was a hot plasma containing photons, electrons, nuclei, and other particles. Before recombination, photons scattered frequently and the universe was opaque. After roughly 380,000 years they free-streamed, producing the cosmic microwave background observed today [19]. The first stars appeared hundreds of millions of years later and produced the first stellar ultraviolet and optical light [30].

A pre-geometric state is a different question. If there is no Lorentzian metric or effective characteristic equation, then c may be **undefined**. Calling it infinite would assert a measurable speed without defining distance, duration, rods, clocks, or a cross-phase comparison.

3.2 Infinite extent does not require infinite speed

A spatially flat FLRW geometry can have infinite constant-time slices and finite local c :

$$ds^2 = -c^2 dt^2 + a(t)^2 [d\chi^2 + \chi^2 d\Omega^2].$$

For a radial light ray,

$$d\chi/dt = \pm c/a(t).$$

At finite cosmic age, causal contact is limited by the integral

$$\chi_{\text{particle}} = \int_0^t c \, dt' / a(t').$$

An infinite spatial slice therefore contains regions outside causal contact. Light does not need to traverse the entire slice “instantly”; there is no outer edge it must reach. Distant recession speeds can exceed c because proper distance changes with the scale factor, not because an object locally passes an observer faster than the causal speed.

The project’s process interpretation of infinity remains philosophically possible: a genealogy can be unbounded even if each causal domain has finite accessible entropy. It is not forced by the finiteness of c .

3.3 Finite-domain causal calibration and the VSL branch

The statement “each space has its own start, end, and light speed” contains three logically different propositions. A domain may have a finite proper-time interval or a transition boundary; a global solution may specify its geometry end to end; and its fields may possess characteristic cones. None of those statements alone implies the others.

For a domain-indexed homogeneous metric with lapse N_i ,

$$ds_i^2 = -N_i(t)^2 c_{\text{ref}}^2 dt^2 + a_i(t)^2 d\chi_i^2,$$

a radial null curve satisfies

$$d\chi_i/dt = \pm N_i c_{\text{ref}}/a_i.$$

That coordinate slope can grow when the scale factor is smaller or the lapse is larger. The comoving observer instead measures

$$\begin{aligned} d\tau_i &= N_i dt, \\ d\ell_i &= a_i d\chi_i, \\ d\ell_i/d\tau_i &= c_{\text{ref}}. \end{aligned}$$

Thus “space becomes larger and thinner” can change horizons, redshift, coordinate crossing times, density, and the number of interactions without changing the locally measured common Lorentzian cone. A smooth positive lapse is locally absorbed into $\tau_i = \int N_i dt$. If a lapse vanishes, a metric degenerates, or a proper-time range ends, that is a distinct boundary or singularity question rather than a variable-light-speed result.

The physical DSF-3 question starts only when an action derives two different principal symbols,

$$\begin{aligned} G_a^{\mu\nu} k_\mu k_\nu &= 0, \\ G_b^{\mu\nu} k_\mu k_\nu &= 0, \end{aligned}$$

and hence a dimensionless same-frame observable

$$\delta_{ab}(z, k, n) = c_a(z, k, n)/c_b(z, k, n) - 1.$$

To compare domains i and j , the global transition must additionally map observers, proper times, phases, sectors, energies, directions, and state normalizations. Only then is

$$R_{ab}(i \rightarrow j) = [(c_a/c_b)_j]/[(c_a/c_b)_i]$$

defined. A bare c_i/c_j is unit dependent. In the absence of this map the correct word is **uncompared**, not faster, slower, or infinite. This is the causal counterpart of NGS proper-frequency transfer.

VSL is a real family of theoretical programmes rather than one mechanism. Albrecht and Magueijo investigated modified cosmological equations with faster early light as an alternative route to the horizon, flatness, and cosmological-constant puzzles [66]. Later work distinguishes hard Lorentz breaking, bimetric models, locally

Lorentz-invariant constructions, frequency-dependent propagation, extra-dimensional mechanisms, and vacuum-polarization effects [67,68]. These theories show that the question can be posed mathematically. They do not derive the monotonic rule “more density or complexity means slower local vacuum light.” Each class must state which cone changes, what symmetry or preferred frame it uses, and which dimensionless observable changes.

The vacuum-medium analogy is similarly conditional. The Lorentz-invariant Minkowski QED vacuum is not a generic material whose refractive index is set by cosmic complexity. In a specified curved background, one-loop vacuum polarization can modify photon characteristics in a curvature- and polarization-dependent way [69]. Specified Casimir boundaries also produce a tiny direction-dependent low-energy refractive correction [70]. Those results justify deriving an effective photon metric from an action; they do not supply a universal density law, and phase or group velocity cannot automatically be identified with a controllable superluminal signal.

A literal $c \rightarrow \text{infinity}$ is not needed. In the fixed-coordinate normalization $ds^2 = -c^2 dt^2 + dx^2$, it sends $g^{(00)}$ toward zero and is a singular Lorentzian-to-Galilean-type limit rather than another regular finite domain. FGC suggests the sharper alternative: a coarser regime may have a **large but finite relative cone**, if a healthy action derives it and if hyperbolicity, stability, EFT control, causal-loop avoidance, local recovery, and observations permit it. This replaces completed infinite speed with a finite value along an unbounded hierarchy, consistently with the project’s treatment of infinity.

Penrose’s Conformal Cyclic Cosmology is a valuable neighboring construction. CCC proposes that the future conformal infinity of one aeon joins the conformally stretched Big Bang of the next [65]. It does not follow that an ordinary comoving observer reaches that future in finite proper time, that an empty universe automatically creates the next aeon, or that sequential aeons are the same geometry as FGC’s nested black-hole domains. A finite conformal representation is not by itself a physical end of time. Recursive Horizons must derive its own time range, transition surface, conformal/physical metric relation, flux, entropy, and causal map.

Finally, quantum entanglement cannot serve as a shortcut. Entanglement violates local-hidden-variable factorization but does not provide selectable faster-than-light messages. For a local trace-preserving operation E_A ,

$$\rho_{B'} = \text{Tr}_A[(E_A \otimes I_B)(\rho_{AB})] = \rho_B.$$

The remote marginal is unchanged, and a classical causal channel is needed to compare outcomes [71]. A proposed top-layer mechanism would therefore require new dynamics that confront no-signalling, Lorentz covariance or a declared preferred foliation, causal loops, and Bell/quantum-information experiments. Entanglement alone is not evidence for such a layer.

DSF-3 is intentionally documentation, not a new result artifact. The existing DSF-2 disformal cone and GW170817 gate already execute the minimum within-domain relative-cone calculation. New code is justified only after a transition action fixes new characteristic inputs and a mapped observable.

3.4 What classical black-hole geometry establishes

In Schwarzschild coordinates, an outgoing radial null curve satisfies

$$dr/dt = c (1 - r_s/r).$$

The coordinate value approaches zero at $r=r_s$. That is not a local photon slowing to rest. Schwarzschild time is singular at the horizon; regular horizon-crossing coordinates pass through it smoothly [8]. The invariant content is causal: once inside the event horizon, all future-directed causal paths in the classical solution move toward smaller areal radius and cannot reach the exterior.

Classical black-hole solutions demonstrate that Lorentzian geometry can contain one-way causal boundaries. They do not establish that local c varies, that a horizon tears into another universe, or that we inhabit a parent black hole.

Nor does the present spherical mechanism test assume that crossing a horizon destroys neutrons or any other observed particle. It asks only whether classical localization of the independent χ_i sector activates the declared geometric/regulator response while the continuum equations, constraints, and characteristics remain healthy. Any claim

that familiar particle degrees are reorganized across a deeper phase boundary belongs to the later NGS spectral problem and requires its own quantum transition map.

3.5 The Planck-scale localization argument

As a heuristic dimensional estimate, equating a quantum localization scale with a gravitational radius gives an instructive order-of-magnitude crossing. Using the reduced Compton wavelength and Schwarzschild radius,

$$\hbar/(m c) = 2 G m/c^2,$$

gives a Planckian mass and length up to factors of order unity. This motivates the expectation that ordinary quantum mechanics on a fixed spacetime and classical general relativity cannot both remain unchanged at arbitrarily short distances [18]. It does not establish an experimentally measured hard pixel of exactly $\sqrt{2} \ell_P$, nor prove that literal micro black holes form under every attempt at Planck-scale measurement. Convention choices change the numerical factor; quantum gravity is precisely what is missing.

4. What general relativity actually contributes

4.1 A black-hole interior is cosmological in a precise sense

For $r < r_s$, the classical Schwarzschild coordinate r is timelike in character and the Schwarzschild interior can be written in a Kantowski–Sachs form [9]. It is a homogeneous, anisotropic geometry with one radial and two angular scale factors. In that limited geometric sense, an interior is a contracting cosmological spacetime rather than an ordinary static room.

This is a substantial clue, but three inferences do not follow:

1. a vacuum eternal Schwarzschild interior is not a realistic rotating stellar-collapse interior;
2. the coordinate/geometric correspondence does not remove the classical singularity;
3. a contracting Kantowski–Sachs branch does not automatically connect to an expanding FLRW child.

The child-universe statement is the proposed extension, not the coordinate transformation.

4.2 The Hubble-sphere compactness identity

For a flat FLRW model, define the critical density and Hubble radius as

$$\rho_{\text{crit}} = 3 H^2/(8 \pi G), \quad R_H = c/H.$$

The critical mass-energy inside a sphere of radius R_H is

$$M_H = (4 \pi/3) \rho_{\text{crit}} R_H^3 = c^3/(2 G H).$$

Therefore

$$C_H = 2 G M_H/(c^2 R_H) = 1.$$

This equality is exact because the same definitions and Friedmann relation appear on both sides. It is not a measured coincidence and it is not sufficient evidence that the universe is a Schwarzschild black hole. The Hubble radius is not generally an event horizon, the observable-universe radius is much larger than c/H_0 , and FLRW lacks the asymptotically flat exterior assumed by the Schwarzschild compactness criterion. In spherical cosmology the Misner–Sharp mass and apparent-horizon language is more appropriate.

The identity remains useful: it explains why horizon-like algebra naturally appears at the cosmological expansion scale. Its correct label is **identity**, not **proof**.

4.3 Why a black hole does not consume every available energy density

Continued collapse does not create infinite total mass. The classical singularity is a statement of geodesic incompleteness and divergent curvature in an extrapolated solution, not a material object containing an infinite amount of mass. The exterior remains characterized by finite conserved or asymptotic charges such as mass, angular momentum, and charge. Far-away matter can orbit, escape, join other bound systems, or remain outside causal contact.

A black hole grows when energy-momentum or gravitational radiation actually crosses the relevant horizon. Dynamical-horizon balance laws relate horizon change to those fluxes [45]. “Energy exists in the surrounding space” is not a flux calculation: trajectories, angular momentum, pressure, interactions, boundary conditions, and the global geometry decide what crosses. A cosmological-constant stress tensor,

$$T_{\mu\nu}^{\Lambda} = -\rho_{\Lambda} g_{\mu\nu},$$

is not an ordinary gas of particles waiting to fall radially inward.

The user-motivated balance intuition nevertheless has a clean established example. The weak-field Schwarzschild–de Sitter/Kottler radial acceleration is [44]

$$a_r = -G M/r^2 + \Lambda c^2 r/3.$$

The compact-mass attraction and positive- Λ acceleration are equal at

$$r_{\text{balance}} = [3 G M/(\Lambda c^2)]^{1/3}.$$

In FLRW language the same sign is visible because vacuum energy has $p=-\rho$, making $\rho+3p=-2\rho$ in the acceleration equation. Attraction and accelerated separation can therefore coexist and define a finite scale within ordinary GR. The repository verifies the Kottler balance algebra directly.

The equality is not a stable equilibrium. At the balance radius,

$$d a_r/dr = \Lambda c^2 > 0,$$

so a small outward displacement accelerates farther outward and a small inward displacement accelerates farther inward. It is an unstable radial separator.

That radius is not a material wall and does not stop ordinary accretion near a horizon. It also does not prove that negative pressure creates space or that a parent domain restrains a child. It does identify the stronger calculation the proposal owes: find a global, horizon-regular solution in which competing stress-energy and geometric terms actually produce a stable or metastable causal domain, then show which perturbation drives it from a trapped branch to an expanding one. This **domain-balance hypothesis** is the proposed bridge; the Kottler equation is its established local precedent.

5. Horizon thermodynamics and the conditional relation

5.1 Established area laws

Semiclassical black-hole thermodynamics assigns entropy

$$S_{\text{BH}}/k_B = A_{\text{BH}}/(4 \ell_P^2).$$

For a spherical Schwarzschild horizon, $A_{\text{BH}}=4 \pi r_s^2$, so

$$S_{BH}/k_B = \pi r_s^2/l_P^2.$$

Bekenstein introduced black-hole entropy and Hawking's particle-creation calculation fixed its temperature and normalization [1,2]. For de Sitter spacetime, Gibbons and Hawking obtained a cosmological horizon radius and entropy [3],

$$\begin{aligned} R_{dS} &= \sqrt{3/\Lambda}, \\ S_{dS}/k_B &= \pi R_{dS}^2/l_P^2 \\ &= 3\pi/(\Lambda l_P^2). \end{aligned}$$

Covariant entropy bounds apply to appropriate non-expanding null light-sheets, not indiscriminately to every spatial volume [5,6]. Island/Page-curve calculations provide major evidence for unitary entropy evolution in controlled semiclassical settings, but they do not specify a black-hole-to-child transition map [7].

5.2 The new saturation postulate

Recursive Horizons may adopt the following strong postulate:

Horizon-saturation postulate (H-SAT). *For a successful parent-to-child transition, the accessible asymptotic de Sitter entropy budget of the child saturates the Bekenstein–Hawking entropy assigned to the parent horizon.*

Symbolically,

$$S_{dS,child} = S_{BH,parent}.$$

This equality is not supplied by unitarity. Fine-grained von Neumann entropy, thermodynamic coarse-grained entropy, horizon entropy, and a Hilbert-space dimension are related but not interchangeable. H-SAT is a boundary condition to derive or test.

5.3 Conditional derivation

For spherical horizons, H-SAT gives

$$3\pi/(\Lambda l_P^2) = \pi r_s^2/l_P^2.$$

Cancelling the common factors yields

$$\Lambda = 3/r_s^2.$$

Using $r_s = 2 G M_{parent}/c^2$,

$$\Lambda = 3 c^4/(4 G^2 M_{parent}^2).$$

This derivation is exact **conditional on H-SAT and the semiclassical horizon formulas being the correct quantities to match**. It does not require modifying local general relativity inside the child's tested regime.

5.4 What the numerical example means

Using rounded Planck 2018 base- Λ CDM values $H_0=67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m \approx 0.315$, and therefore $\Omega_\Lambda \approx 0.685$ in a flat late-time approximation [19], one obtains

$\Lambda = 3 H_0^2 \Omega_\Lambda / c^2$	approximately $1.09e-52 \text{ m}^{-2}$,
$R_{dS} = \sqrt{3/\Lambda}$	approximately $1.66e26 \text{ m}$,
$M_{inferred} = c^2 R_{dS}/(2G)$	approximately $1.12e53 \text{ kg}$,
S_{dS}/k_B	approximately $3.3e122$.

The parent mass is labeled **inferred** because Λ determined it. If an assumed parent mass is first chosen to equal this value and its formula then returns the same Λ , the agreement is circular. The calculation shows that H-SAT maps the observed cosmological scale to a parent horizon of comparable order to a Hubble-scale mass; it does not show that such a parent exists.

The same caution applies to ratios such as $R_H/R_D = \sqrt{\Omega_\Lambda}$. That is a valid Λ CDM algebraic relation. Calling it “83% of a radial journey” adds an operational interpretation not established by the equation.

5.5 Rotation

A Kerr horizon has a spin-dependent area, so a rotating version of H-SAT would change the inferred mass-radius relation. That is a legitimate conditional extension. It does not permit spin to be estimated a second time from an invalid CMB statistic and then called independently confirmed.

A rotating model must derive:

- a regular interior/transition geometry;
- which angular momentum is conserved and in what global spacetime;
- how coherent rotation, shear, vector perturbations, and inflation interact;
- a signed observable template and transfer function.

For version 0.11.0, rotation is an open module rather than part of the minimal core.

6. Candidate transition mechanisms

6.1 Einstein–Cartan gravity

Einstein–Cartan–Sciama–Kibble gravity extends the connection to include torsion sourced by spin [10,11]. In the minimal theory torsion is algebraic and does not propagate as an independent vacuum wave. Eliminating it can produce an effective local four-fermion interaction. At high spin density this contribution can alter collapse dynamics.

Microscopic and spin-fluid cosmological models show nonsingular bounces under specified assumptions [12]. Poplawski has developed directly relevant black-hole-universe models in which torsion, particle production, and a closed anisotropic interior avoid a singularity [13]. These are valuable existence candidates.

The remaining conditions are decisive:

- the fermionic state and spin correlations must be defined;
- averaging must remain valid near the transition;
- shear must not dominate the torsion term;
- particle production and backreaction must be controlled;
- a realistic collapse exterior and junction must be supplied;
- perturbations must remain stable and recover observation.

It is therefore accurate to say “ECKS models can bounce.” It is not accurate to say “ECKS proves every black hole bounces.”

6.2 Loop quantum cosmology

In improved-dynamics LQC, a spatially flat homogeneous isotropic model with a massless scalar undergoes a deterministic quantum bounce at a Planckian density [14]. Effective equations are often written schematically as

$$H^2 = (8\pi G/3) \rho (1 - \rho/\rho_c).$$

This result is powerful within its model. It does not automatically apply to an astrophysical black-hole interior, and it does not validate an Einstein–Cartan mechanism. The two frameworks may offer comparative boundary conditions; their rough density scales are not an “overdetermined proof.”

6.3 Tunnelling and baby universes

Semiclassical work has explored whether a new inflating region can be connected by quantum tunnelling [15]. Such calculations show that “child universe” is a real theoretical research category, not an incoherent phrase. They also demonstrate why topology, boundary conditions, negative modes, causal embedding, and probability amplitudes cannot be skipped.

6.4 The transition is the theory

The black-hole exterior, the collapsing interior, and the expanding child are not unified merely by drawing a tunnel. The transition needs a solution or controlled quantum amplitude. At minimum it must connect a trapped region to an expanding branch containing the appropriate anti-trapped region while keeping the effective description finite:

```
trapped region -> finite transition -> expanding branch with anti-trapped region.
```

An expanding cosmology is not anti-trapped everywhere: with a standard future-directed null-normal convention, normal spheres occur on one side of an apparent horizon and anti-trapped spheres on the other. The signs, normalization, horizon locations, and global interpretation must be supplied by the actual solution. That invariant expansion structure is a much better target than “space turns inside out.”

6.5 CCT-1: a controlled covariant core

Version 0.3 introduces one deliberately conservative realization, **CCT-1**. In units $c=\hbar=1$, with signature $(-, +, +, +)$ and reduced Planck mass $M_{\text{Pl}}^2 = (8\pi G)^{-1}$, its action is

```
S_CCT1 = integral d^4x sqrt(-g)
[ M_Pl^2 R/2
  - (nabla Phi)^2/2
  - V(Phi) ]
+ S_matter.
```

Φ has mass dimension one and V has mass dimension four. The equations are second order. For the exact solution below, ordinary matter is absent: the stress tensor from S_{matter} is set to zero, with any vacuum-energy contribution absorbed into V_0 . Radiation, dust, or a generic fluid would change that solution. At a positive-potential stationary point,

```
V, Phi(Phi_0) = 0,
V,_{Phi Phi}(Phi_0) = m_Phi^2 >= 0,
V(Phi_0) = V_0 > 0,
L^2 = 3 M_Pl^2/V_0.
```

$m_{\text{Phi}}^2 > 0$ gives a strict quadratic minimum. The allowed equality $m_{\text{Phi}}^2 = 0$ establishes only linear non-tachyonicity of the potential; higher derivatives must then determine stability.

On closed FLRW slices,

```
ds^2 = -dt^2 + a(t)^2 [dchi^2 + sin^2(chi) dOmega_2^2],
```

the exact stationary-scalar solution is [31]

```
Phi(t) = Phi_0,
a(t) = L cosh(t/L),
H(t) = tanh(t/L)/L.
```

The scale factor contracts, reaches $a=L$ at $t=0$, and expands. Its invariants are finite and constant,

$$\begin{aligned} R &= 12/L^2, \\ R_{\mu\nu} R^{\mu\nu} &= 36/L^4, \\ R_{\mu\nu} \rho_{\sigma} \sigma^2 &= 24/L^4. \end{aligned}$$

For the more general canonical-scalar closed equations, a putative bounce with $H_b=0$ obeys

$$\begin{aligned} 1/a_b^2 &= [\text{dot}(\Phi_b)^2/2 + V(\Phi_b)]/(3 M_{Pl}^2), \\ \text{dot}(H_b) &= [V(\Phi_b) - \text{dot}(\Phi_b)^2]/(3 M_{Pl}^2). \end{aligned}$$

Thus positive spatial curvature permits a bounce without a ghost or null-energy-condition violation when $V(\Phi_b) > \text{dot}(\Phi_b)^2$. CCT-1 uses the exactly solvable stationary point as a baseline. A genuinely evolving order parameter would require a new background and a new perturbation calculation rather than silently borrowing the formulas below.

For areal radius $R_A = a \sin(\chi)$, choose future radial null vectors

$$\begin{aligned} k_+ &= \text{partial}_t + a^{-1} \text{partial}_\chi, \\ k_- &= \text{partial}_t - a^{-1} \text{partial}_\chi, \\ k_+ \cdot k_- &= -2. \end{aligned}$$

The expansions are then

$$\begin{aligned} \theta_{+} &= 2 [H + \cot(\chi)/a], \\ \theta_{-} &= 2 [H - \cot(\chi)/a]. \end{aligned}$$

Their product vanishes on $|aH| = |\cot(\chi)|$. Early enough in the contracting branch there are future-trapped spheres; at the bounce, non-equatorial spheres are normal; late enough in the expanding branch there are future-anti-trapped spheres. The calculation therefore derives the region structure rather than assigning one sign pair to the whole child.

The limitation is equally concrete. This solution is global de Sitter space in a closed slicing, not a Schwarzschild or Kerr interior joined to an exterior. Its curvature invariants do not mark an invariant microphysical phase change. It establishes a covariant **core laboratory** without a ghost or gradient pathology and fixes a null-expansion convention; it does not establish boundedness of every infrared mode or satisfy the still-open global embedding, topology, junction, shear, reheating, or hot-Big-Bang requirements. If its curvature approaches the EFT cutoff, the two-derivative action also ceases to justify itself. Stable flat bounces instead require carefully specified NEC-violating or modified-gravity dynamics; broad Horndeski constructions face a no-go result, while model-specific beyond-Horndeski/DHOST examples provide possible—not automatic—loopholes [37,38,39].

6.5.1 GMF-1: a direct-match obstruction, not a child-universe construction

The right first global calculation is allowed to fail. Oppenheimer and Snyder provide an exact control in which a finite closed dust-FLRW interior is matched smoothly to Schwarzschild [49]. It shows that finite mass and a smooth FLRW/Schwarzschild junction are compatible in classical GR. It also reaches a classical singularity, so it is not a regular bounce or a child universe.

For the proposed direct attachment to the CCT-1 core, spherical quasi-local mass supplies an exact obstruction. With areal radius R , the Misner–Sharp mass is

$$\begin{aligned} M_{MS, \text{Kottler}} &= M + \Lambda R^3/(6 G), \\ M_{MS, \text{CCT-1}} &= \Lambda R^3/(6 G). \end{aligned}$$

At equal Λ , a finite $M > 0$ therefore leaves $\Delta M_{MS} = M$. A no-shell Darmois junction requires compatible induced metric and extrinsic curvature, and this direct Kottler-to-pure-CCT-1 bridge cannot meet that requirement [48,50]. This is a restricted result: it does not eliminate an inhomogeneous transition layer, a specified shell, or modified gravity. It does eliminate the claim that the two displayed metrics alone form a smooth finite-mass parent-to-core solution.

An Israel shell can be prescribed to carry the discontinuity [50]. That gives distributional local interface data, not a completed transition: its stress, conservation/flux law, energy conditions, invariant regularity, radial and bulk stability, and global causal completion must still be derived. A null interface uses the distinct Barrabes–Israel formalism [51]. Likewise, the parent and CCT expansion pairs are local quantities in different geometries; they become comparable only in a single horizon-regular global solution with one declared orientation and matching prescription. The spherical unified first law is the appropriate framework for the missing quasi-local energy balance [52].

The possible change from finite-mass collapse Cauchy data to a compact closed child also cannot be assumed away. Smooth classical topology-change statements require their causal hypotheses to be checked explicitly; under Geroch's relevant conditions topology change entails causal pathology [54]. Separately, a trapped surface plus the null-convergence/global assumptions of Penrose's theorem produces null geodesic incompleteness [53]. These theorems do not prove that every quantum-gravity completion fails; they identify assumptions a successful regular global model must state and, where needed, evade or replace.

The canonical GMF-1 record is documented at [docs/global-matching-flux-closure.md](#) and reproduced by `python3 scripts/reproduce_gmf.py`. It contains no derived outside-dark-energy mechanism and no varying-c result.

6.5.2 EC-1: a homogeneous finite cap is a candidate, not GMF-1B

Einstein–Cartan spin-fluid collapse supplies a useful action-motivated reduced candidate rather than a shell prescription. After the stated homogeneous/isotropic spin averaging, its closed finite-cap Friedmann equation can be written

$$\dot{a}^2 = -1 + A/a^2 - B/a^4, \quad A > 0, B > 0.$$

The positive energy-density contribution is radiation-like and proportional to a^{-4} ; the torsion effective contribution is negative and proportional to a^{-6} . The separately conserved fermion number density, when invoked in the spin average, is proportional to a^{-3} . The torsion term has effective equation of state $w=+1$, not $w=-1$. If $A^2 > 4B$, the two turning radii are

$$a_{\pm}^2 = [A \pm \sqrt{A^2 - 4B}]/2,$$

and contraction reaches the finite lower turning point a_- , where $\ddot{a} > 0$, before re-expanding. Consequently the homogeneous curvature and stress scalars built from this reduced solution remain finite at that turning point. This is a local finite-cap result, not a Planck-complete calculation. It follows the Einstein–Cartan/Weyssenhoff reductions studied in [55,56].

Using one closed-FLRW interior metric and the already declared future null normals, the same solution can contain a trapped phase during contraction, a normal non-equatorial neighbourhood at its finite bounce, and anti-trapped spheres during expansion. That is stronger than juxtaposing independent parent and child sign samples, but it is still an **interior** causal sequence. It does not construct a finite-mass exterior, establish a global topology, or identify our universe with the expanding branch.

Its sharpest negative result is at the boundary. For a finite cap at fixed χ_b , $R_b = a \sin(\chi_b)$, and the FLRW Misner–Sharp boundary mass obeys

$$M_{MS,b} = (4\pi/3) \rho_{eff} R_b^3, \\ \dot{M}_{MS,b} = -4\pi R_b^2 p_{eff} \dot{R}_b.$$

The negative torsion component has $p_{tor} = \rho_{tor}$ and nonzero pressure, so $M_{MS,b}$ drifts through both contraction and expansion. A fixed-mass static Schwarzschild exterior cannot therefore remain smoothly matched to this homogeneous pressureful boundary. A physically completed model needs a resolved pressure taper, an exterior energy flux, or a layer with its own conservation law. None of those is supplied here. Thus EC-1 makes no derived Q , dark-energy, relative-cone, or varying-c claim, and it leaves the full horizon-regular 1+1 Einstein–Cartan–Dirac GMF-1B problem open.

6.5.3 GMF-1B-PF1: preflight gates reject two shortcut closures

The next calculation cannot simply insert a familiar bounce ingredient into a radial solver. It must have both the required defocusing mechanism and a regular causal principal part. First consider a smooth canonical scalar, minimally coupled to ordinary Einstein gravity. For every metric-null vector k^a , its stress tensor obeys

$$\begin{aligned} T_{ab} k^a k^b &= (k^a \nabla_a \Phi)^2 \geq 0, \\ R_{ab} k^a k^b &= 8 \pi G T_{ab} k^a k^b \geq 0. \end{aligned}$$

For the twist-free null congruences relevant to spherical symmetry,

$$d \theta / d \lambda = -\theta^2/2 - \sigma_{ab} \sigma^{ab} - R_{ab} k^a k^b \leq 0.$$

This exact Raychaudhuri gate rejects that *restricted canonical Einstein--scalar action* as the material which reverses a metric-null trapped congruence. It does not turn CCT-1's regular closed de Sitter control into a black-hole transition, and it does not exclude noncanonical fields, modified gravity, torsion/spinor dynamics, or another derived closure [57].

Second, the specific homogeneous thermal EC average used in EC-1 cannot be promoted naively to a causal radial perfect fluid. With $z=B/(A - a^2)$, its exact barotropic derivative is

$$\begin{aligned} c_s^2 = dp/d\rho &= (2-9z)/[3(2-3z)], \\ z_{\text{bounce}} &= [1+\sqrt{1-4B/A^2}]/2, \quad 1/2 < z_{\text{bounce}} < 1. \end{aligned}$$

The thresholds are $c_s^2=0$ at $z=2/9$, $\rho+p=0$ and a pole at $z=2/3$, $\rho+3p=0$ at $z=1/2$, and $\rho=0$ at $z=1$. Therefore the route from the radiation-like regime to the lower bounce necessarily crosses $c_s^2<0$, and where it continues through $z=2/3$ it also reaches a pole and then $c_s^2>1$. This rejects the stated averaged T^4 - T^6 **perfect-fluid closure** as a causal/hyperbolic GMF-1B radial material. The derivative is an adiabatic-fluid diagnostic, not a microscopic EC characteristic calculation; the result consequently leaves full EC--Dirac and other action-derived tapered closures open [55,56].

The constructive geometry is an expanding finite B^3 cap, not an inferred complete S^3 child. Under smooth connected globally hyperbolic evolution with fixed topology, a noncompact finite-mass parent Cauchy slice cannot become a later complete boundaryless S^3 Cauchy slice. A full-child proposal must state which premise changes, rather than hiding a topology change in a local cap [54,58,59]. If a candidate develops an inner/Cauchy horizon, it has one further gate: it must show either that such a horizon is absent or that counter-streaming perturbations keep quasi-local mass and curvature bounded under convergence refinement. The mass-inflation literature makes this a required stability calculation, not a universal proof that every bounce fails [60].

The deterministic restricted-gate record is documented in `docs/gmf-1b-preflight.md` and reproduced by `python3 scripts/reproduce_gmf1b_preflight.py`. It contains no 1+1 solution, no global junction, and no observation-level prediction.

6.5.4 GMF-1B-ECD-SYM1: a viable axial-current sector, not yet a full source or spacetime

PF1 rejects a canonical Einstein--scalar defocusing route and the naive thermal EC perfect-fluid material law; it does not reject minimal Einstein--Cartan--Dirac theory. The next question is therefore narrower than a bounce simulation: can a rotationally compatible spinor source retain the minimal ECD axial channel at all?

SYM1 uses the Ventrella--Choptuik numerical representation with signature $(-, +, +, +)$, $\epsilon_{\text{hat}0123}=+1$, $\kappa=8 \pi G$, and $\gamma_{\text{star}}=i \gamma_{VC^0} \gamma_{VC^1} \gamma_{VC^2} \gamma_{VC^3}$. In this imaginary gamma representation the real axial current used by the executable is

$$\begin{aligned} A^a &= -i \sum_I \bar{\psi}_I \gamma_{VC^a} \gamma_{\text{star}} \psi_I, \\ \bar{\psi}(\psi) &= \psi^\dagger \gamma_{VC^0}. \end{aligned}$$

The canonical ECD action and its translation to this representation are fixed and tested separately by INT1 below. In the VC convention the resulting full torsion-eliminated contact interaction is $L_{\text{contact}}=3 \kappa A_a A^a/16$ [61,64]. The SYM1 artifact compares two spherical constructions that must not be conflated. Its declared equal-profile,

opposite-spin full-Dirac singlet control has zero total axial current, so its minimal axial torsion channel cancels even though its ordinary Einstein–Dirac stress need not vanish [63]. The Ventrella–Choptuik massless left-chiral pair is different [62]. Direct gamma-matrix evaluation gives, in the unit-normalization algebraic fixture,

$$\begin{aligned} A^\theta &= 2 (|F|^2 + |G|^2), \\ A^r &= 2 (|F|^2 - |G|^2), \\ A_a A^a &= -16 |F|^2 |G|^2. \end{aligned}$$

Thus the axial contact source is active whenever both radial amplitudes are nonzero. At $F=3/4$, $G=1/2$, $\kappa=1$, the exact binary-rational checks are $A^\theta=13/8$, $A^r=5/8$, $A^2=-9/4$, and $L_{\text{contact}}=-27/64$. Restoring the polar-areal spinor normalization multiplies the current by the appropriate powers of $C=1/(2 \sqrt{\pi} r \sqrt{a})$; the unit fixture is not a physical density at an unspecified geometry.

The one-amplitude controls require a separate distinction. When $F=0$, $G \neq 0$ or $G=0$, $F \neq 0$, the current is nonzero and null: it still sources algebraic Cartan torsion, while $A_a A^a=0$ makes the scalar contact-action density vanish at that field value. A zero invariant does not generally determine the varied Hehl–Datta term. For this declared VC pair, INT1 directly verifies that the projected local cubic contact source also vanishes on either one-amplitude branch.

The total axial-current sector is compatible with **SO(3)**: temporal and radial orbit-space components are allowed while angular components vanish. A radial axial vector can break parity, so no $O(3)$ -invariance claim is made. The complete nonlinear ECD source has not yet been shown $SO(3)$ -equivariant; that requires the full stress and complete harmonic projections below. The fields are classical complex c-number fields, not a derived quantum two-fermion state, Fermi gas, or Weyssenhoff ensemble.

The four-fermion interaction contains no derivatives. It changes nonlinear sources and the effective stress but not the minimally coupled Dirac principal symbol,

$$\det P(\xi) \text{ proportional to } (g^{ab} \xi_a \xi_b)^2.$$

The characteristics therefore remain the metric-null cone; SYM1 supplies no variable-local-c or relative-cone prediction. Its normalized C -infinity annular packet is exactly zero in a centre buffer and an exterior buffer and has a finite nonzero axial contact invariant in between. Those are exact spinor-zero buffers, not yet constraint-compatible vacuum spacetime regions, a permanent material wall, or a shell-free exterior match.

The result stops at precisely the important boundary. KIN1 subsequently closes the algebraic contact-sector tetrad variation, but the programme has not derived every torsion-free Dirac stress component, proved complete spin-weighted harmonic closure, derived a regular-centre series, solved the null constraints, or evolved curvature. In particular $A^2 < 0$ is not a proof of negative energy, Raychaudhuri defocusing, or a bounce.

The canonical source record is documented in `docs/gmf-1b-ecd-symmetry.md` and reproduced by `python3 scripts/reproduce_gmf1b-ecd_symmetry.py`. Its interaction successor is GMF-1B-ECD-INT1.

6.5.5 GMF-1B-ECD-INT1: exact interaction bridge, not a complete ECD source

INT1 first defines the physical minimal-ECD matter interaction in a canonical $(+, -, -, -)$ convention, then translates the complete matter equation to the Ventrella–Choptuik $(-, +, +, +)$ gamma representation. The two convention layers are related by

$$\begin{aligned} \Gamma_a A^a &= -i \gamma_a \gamma^a, \\ \Gamma_a A^5 &= i \Gamma_a A^\theta \Gamma_a A^1 \Gamma_a A^2 \Gamma_a A^3 = \gamma_a \gamma^a, \\ J^a &= A^a = -i \bar{\psi} \gamma^a \gamma^a \psi, \\ J_{+}^2 &= -A_{-}^2. \end{aligned}$$

Cabral, Lobo, and Rubiera-Garcia's effective-action decomposition (their Eqs. 8–11 and 23), spin-vector relation (Eq. 18), and substituted Dirac sector (Eq. 20) combine into the complete torsion-eliminated canonical interaction; their Eq. 83 gives the Hehl–Datta equation. They use $\kappa^2=8 \pi G$ where this paper uses $\kappa=8 \pi G$ [64]. In the notation here,

```

L_int,+ = -(3 kappa/16) J_+^2,
L_int,- = +(3 kappa/16) A_-^2,

gamma_VC^a D_a psi
- i (3 kappa/8) A_a gamma_VC^a gamma_star psi = 0.

```

This is one global convention translation, not two physical metrics. The tetrad, orientation and epsilon, index lowering, curvature/action signs, and stress convention must transform coherently with the signature. INT1 directly checks both Clifford algebras, the common adjoint, the gamma-five relation, the real-current map, the invariant and interaction signs, and the matrix field- equation translation. For $C=1$, $\kappa=1$, $F=3/4$, and $G=1/2$, it obtains

```

J_+^2 = +9/4,      A_-^2 = -9/4,
L_int,+ = L_int,- = -27/64,
maximum signature-bridge residual = 8.88e-16.

```

For the polar-areal normalization $C=1/(2 \sqrt{\pi}) r \sqrt{a}$, sphere reduction gives

```

L_4,red = -(3 kappa alpha)/(4 pi r^2 a) |F|^2 |G|^2,

dL_4,red/dF* = -(3 kappa alpha)/(4 pi r^2 a) |G|^2 F,
dL_4,red/dG* = -(3 kappa alpha)/(4 pi r^2 a) |F|^2 G.

```

The direct cubic operator reconstructs pointwise in the declared local opposite-chirality two-vector basis at three nondegenerate tested angles,

```

N_F = -(3 kappa/2) C^2 |G|^2 F,
N_G = -(3 kappa/2) C^2 |F|^2 G,

dL_4,red/dF* = 2 alpha N_F,
dL_4,red/dG* = 2 alpha N_G.

```

The maximum weak-identity residual is $3.33e-16$; the maximum tested-angle local reconstruction and coefficient residuals are $7.85e-17$ and $1.14e-16$. These equalities compare the covariant interaction operator with variation of the same reduced interaction action. They are not the normalization of a time-evolution equation, an integration against a complete spin-weighted harmonic basis, or a proof that all omitted (ℓ, m) coefficients vanish.

INT1 also records the metric-proportional contact expression $T_{mn}^{\wedge}(4)=(3 \kappa A^2/16) g_{mn}$. KIN1 supplies its tetrad justification: the internal invariant $A^{\wedge}I A_I$ is tetrad independent at fixed spinor components. The contact term's contraction with a metric-null vector therefore vanishes. This is the complete algebraic contact-sector variation, not the torsion-free Dirac kinetic stress or the full ECD source, and supplies no null-defocusing or bounce conclusion.

The deterministic record is documented in `docs/gmf-1b-ecd-identity.md` and reproduced by `python3 scripts/reproduce_gmf1b_ecd_identity.py`.

6.5.6 GMF-1B-ECD-TOR1/KIN1: corrected contact stress, not defocusing

TOR1 independently checks the Hodge-dual spin tensor and Cartan torsion invariant,

```

S^{abc} = (1/2) eps^{abcd} A_d,
T_{abc} T^{abc} = -(3/2) kappa^2 A^2.

```

It imports the full reduced $3 \kappa A^2/16$ contact coefficient from the action after the independent connection has been eliminated everywhere. It does not identify the Einstein–Hilbert torsion term alone with that complete interaction or add convention-dependent intermediate connection pieces to an already-reduced action.

The original KIN1 varied $g^{\wedge mn} J_m J_n$ while holding the coordinate covector J_m fixed. That produces an anisotropic term, but a Dirac axial current is $J_m = e^{\wedge} I_m J_I$ and varies with the coframe. Corrected KIN1 checks a generic non-diagonal coframe in all 16 independent directions and verifies

```
delta(J^2)
= J_m J_n delta(g^mn)
+ 2 g^mn J_m delta(J_n)
= 0.
```

The largest fixed-current partial is about 4.5494; the current response cancels it with maximum residual below $2e-15$. The physical contact stress is therefore

```
T_mn^(4) = L_4 g_mn,
rho = 27/64,
p = -27/64,
rho+p = 0,
T_mn^(4) k^m k^n = 0.
```

The former fixed-current local NEC violation and its isotropic average are retracted. KIN1 also checks one canonical positive-frequency massless plane wave, $p^a=(1,0,0,1)$ and $u=(1,0,1,0)$, for which $\gamma^a p_a u=0$, $j^a=2p^a$, and $T_{ab}=2p_a p_b$. Parallel and opposite null probes give 0 and 8. This is a free-field fixture, not a general classical-Dirac energy-condition theorem.

GMF-1B-ECD-ID1 must now derive the complete torsion-free Dirac kinetic and spin-connection stress/source from this convention-locked action, prove the required harmonic closure, and solve constraint-compatible regular-centre, finite-Misner–Sharp-mass initial data. Only after that passes does a horizon-regular 1+1 evolution become interpretable.

6.6 DST-1: one existing scalar, three established regimes

DST-1 does not add a second mysterious substance. It asks how much of the desired language is already available in the canonical scalar used by CCT-1. Expand its potential around a strict minimum,

```
Phi = Phi_0 + varphi,
V(Phi) = V_0 + m^2 varphi^2/2,
V_0 >= 0, m > 0.
```

For a homogeneous field,

```
rho_Phi = V_0 + dot(varphi)^2/2 + m^2 varphi^2/2,
p_Phi = -V_0 + dot(varphi)^2/2 - m^2 varphi^2/2.
```

For a nonzero constant offset, $V_0>0$, the stationary component is exactly vacuum-like,

```
rho_0=V_0, p_0=-V_0, w_0=-1.
```

For rapid coherent oscillations, $m\gg H$, the quadratic excitation obeys the virial average [41]

```
<dot(varphi)^2> = m^2 <varphi^2>,
<p_osc> = 0,
<w_osc> = 0,
<rho_osc> proportional to a^-3.
```

More generally, a monomial V proportional to $|\varphi|^n$ has

```
<w> = (n-2)/(n+2),
rho proportional to a^[-6n/(n+2)].
```

The repository verifies these identities and an exact fixed- H underdamped spectator endpoint,

```
rho_osc(t_N) a(t_N)^3 exp(Gamma t_N) = rho_osc(0),
```

at integer damped periods. The fixed-H calculation is an analytic spectator benchmark, not a self-consistent Friedmann solution.

If explicit daughter fields give the excitation a decay width Γ , a coarse perturbative transfer ledger can be written

```
Q = Gamma dot(varphi)^2,
dot(rho_osc) + 3H(rho_osc+p_osc) = -Q,
dot(rho_r) + 4H rho_r = +Q.
```

The code verifies that the summed continuity residual vanishes. The action in CCT-1 does not contain the daughter fields or derive Γ ; nonperturbative production, backreaction, and thermalization remain open. A component that reheats efficiently also decays, whereas present-day dark matter must be long-lived. A viable model must explain which excitation transfers energy and which, if any, survives.

This is the concrete content of the “it can be both” intuition. One field can have a vacuum-like background, matter-like coherent excitations, and an interaction-driven radiation channel in different regimes. But background equations do not establish physical identity. To call the excitation dark matter requires its abundance, perturbations, Jeans/sound scale, lensing, halos, and structure growth. To call V_0 dark energy requires the observed scale, radiative control, and late-time likelihood. To call Q the birth of the hot Big Bang requires microscopic couplings, entropy production, thermalization, BBN, and the CMB. DST-1 converts the idea into a staged set of tests rather than declaring those tests passed.

7. The information problem

7.1 A map, not a slogan

A schematic parent-to-child process can be written as

```
V: H_collapse -> H_exterior/radiation tensor H_child.
```

If V is an isometry, it preserves inner products. But gravity complicates factorization: gauge constraints and gravitational dressing obstruct a naive tensor product of spatial regions. A real construction must state the algebras of observables, the approximation in which sectors factorize, and the entropy being compared.

Modern island calculations support unitary Page-curve behavior in controlled settings [7]. They do not prove that information is deposited in a child universe, nor that a parent horizon’s coarse-grained entropy equals a child de Sitter entropy.

7.2 IM-1: an explicit code-subspace isometry

The first information artifact is finite and exact. It works only in a stated semiclassical code-subspace approximation: fix a regular background, a relational dressing/reference prescription, d low-energy dressed code states with negligible backreaction, and an error tolerance. The approximation supplies

```
H_in^code = span{|i> : i=0,...,d-1},
H_R^toy tensor H_C^toy,
dim(H_R^toy) = dim(H_C^toy) = d+1.
```

Each output sector contains the d code labels plus an orthogonal blank state $|perp\rangle$. For $0 \leq p \leq 1$, define

```
V_p |i> = sqrt(1-p) |i>_R |perp>_C
        + sqrt(p) |perp>_R |i>_C.
```

Orthogonality of the two location branches gives

```
V_p^dagger V_p = I_d.
```

Thus $\rho \rightarrow V_p \rho V_p^\dagger$ is a completely positive trace-preserving isometric encoding. Tracing either sector gives complementary flagged erasure channels,

$$\begin{aligned}\rho_R &= (1-p) \rho + p |\text{perp}\rangle\langle\text{perp}|, \\ \rho_C &= p \rho + (1-p) |\text{perp}\rangle\langle\text{perp}|.\end{aligned}$$

The unknown state is coherently located, not copied. For an interior value $0 < p < 1$, neither marginal is a deterministic identity channel; the standard inner-product argument forbids a universal perfect cloner [34]. The repository verifies the matrix isometry, both reductions, trace and Hermiticity preservation, recovery on the code range, and the no-cloning overlap contrast.

This construction uses Stinespring’s channel language [33], but it does not evade the gravitational subsystem problem. Gauge constraints and boundary/edge data obstruct a naive exact factorization by spatial region [35]. H_R^{toy} tensor H_C^{toy} is therefore a controlled interface assumption, not a derived factorization of quantum gravity and not a dynamical amplitude from the CCT-1 action.

The entropy ledger remains explicit. With natural logarithms and $h_2(p) = -p \ln p - (1-p) \ln(1-p)$, isometry preserves global fine-grained entropy, while

$$\begin{aligned}S(R) &= h_2(p) + (1-p) S_{\text{vN}}(\rho), \\ S(C) &= h_2(p) + p S_{\text{vN}}(\rho).\end{aligned}$$

For a pure input the full $R+C$ output is pure even when both marginals are mixed. Generalized entropy instead adds a renormalized bulk entropy to an area term on a specified surface [36]. Bekenstein–Hawking and de Sitter entropies are semiclassical coarse geometric quantities. None of these objects may be substituted for another without a physical derivation.

There is no logical implication in either direction:

$$\begin{array}{lll}V_p^\dagger V_p = I & \text{does not imply} & S_{\text{BH,parent}} = S_{\text{dS,child}}, \\ \text{H-SAT} & \text{does not imply} & \text{a channel or an isometry.}\end{array}$$

IM-1 therefore completes a mathematical demonstration, not RH-3. RH-3 still requires relational observables, constraint-compatible dressing, an error-controlled factorization, backreaction, and dynamics that derive p or replace this toy map.

7.3 Inheritance of constants

The historical corpus proposed that children inherit dimensionless constants with small mutations. This idea resembles cosmological natural selection, but the former mutation law was dimensional speculation and its fertility scan did not support a local maximum at the observed point.

Any inheritance law must act on dimensionless couplings or ratios, not unit-dependent numbers such as “the numerical value of c in metres per second.” It must be derived from V or another transition dynamics, define a probability measure, and predict a distribution that can be compared without anthropic post-selection.

For the minimal model, inheritance means only that a transition map may carry correlations. The form and magnitude are open.

8. Observational reality in 2026

8.1 The standard cosmological baseline

Planck 2018 found that the six-parameter spatially flat Λ CDM model gives a strong description of temperature, polarization, and lensing data. In that model it reported approximately $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315 \pm 0.007$, and $n_s = 0.965 \pm 0.004$ [19,21]. These values are inferences within the model, not direct proof of H-SAT or a parent domain.

A later analysis of the final PR4/NPIPE release obtained Λ CDM constraints consistent with the Planck 2018 baseline and reported smaller uncertainties for several parameters in its likelihood combination [22]. This is a refinement of the standard-model inference, not evidence for a recursive transition.

The model developed here must recover at least the same acoustic structure, adiabaticity, near scale invariance, BBN, lensing, and structure growth before a small anomaly can be credited to the transition.

8.2 DESI DR2

The 2025 DESI paper arXiv:2503.14738 reports DR2 **BAO** measurements and cosmological constraints, not a DR2 full-shape analysis [25]. DESI BAO are well described by flat Λ CDM, while BAO and CMB parameters show a mild reported tension. A time-varying w_0 - w_a model is preferred over Λ CDM at 3.1σ for DESI BAO+CMB and at 2.8 – 4.2σ when different supernova samples are added.

This does not establish evolving dark energy, but it means a strict inherited constant- Λ branch is under legitimate pressure. At the same time, the paper's four published constant- w marginal summaries each place $w=-1$ inside an approximate mean plus or minus 1.96σ interval. There is no contradiction: the time-varying model comparison and the one-parameter constant- w summaries answer different conditioned questions. The correct posture is “test ongoing,” not “passes because significance is below an arbitrarily chosen five-sigma kill line.” Model comparison, systematics, dataset choice, and future observations matter.

Version 0.6 also evaluates the collaboration's public 13-observable BAO vector and full covariance directly. In the declared flat late-time constant- w profile, with $A=c/(H_0 \cdot r_d)$ profiled analytically, the result is

```
w_best      = -0.911899
Omega_m,best = 0.297693
H0*r_d      = 9975.091 km/s
chi2_min    = 9.0410      (10 nominal degrees of freedom).
```

The one-parameter profile intervals are $[-0.9911, -0.8361]$ at the conventional $\Delta \chi^2=1$ threshold and $[-1.0710, -0.7659]$ at $\Delta \chi^2=3.8415$. The calculation omits radiation at sub-percent late-time accuracy and does not include CMB, supernovae, or full-shape information. It therefore checks a real BAO background likelihood without replacing the collaboration's complete analysis or identifying what physical sector generated the fitted expansion.

8.3 Planck isotropy and directional spectral index

Planck 2018 documented several large-angle temperature features while finding no unambiguous corresponding polarization evidence [20]. A later PR4 hemispherical-power study reported a notable temperature signal, but its polarization result was mask sensitive and not robust enough for a firm detection [23].

The most direct 2025 PR4 regional-parameter arXiv preprint fitted cosmological parameters in sky patches and compared dipoles with 600 end-to-end simulations [24]. Its fitted n_s dipole was consistent with isotropy: probability-to-exceed values were 38.8% for temperature alone and 5.8% for temperature plus polarization, with analysis variants larger than the latter. The potentially more unusual parameter dipole was A_s , not n_s .

It is therefore incorrect to say Planck PR4 discovered a direction-dependent scalar tilt. The anomalies justify carefully frozen future tests; they do not confirm a parent Kerr axis.

8.4 Local causal-speed constraints

GW170817 and GRB 170817A constrained the fractional difference between gravitational-wave and electromagnetic propagation speeds along the observed path to roughly $-3e-15$ to $+7e-16$, subject to emission-delay assumptions [26]. Any effective model in the accessible late universe must respect this extraordinary agreement.

The repository now validates exact-hash GWOSC event metadata, the Fermi/GBM trigger FITS, the LVC notice, and GCN Circular 21520. From the paper's declared inputs it reconstructs

```
t_gamma,onset,geocenter - t_GW,merger,geocenter = 1.737774 s,
```

consistent with the published $+1.74 \pm 0.05$ s. The upper speed bound assumes simultaneous first-photon/GW emission; the lower bound allows the gamma-ray signal to be emitted ten seconds later. This validates source provenance and the published arithmetic. It is not a raw strain/light-curve reanalysis and cannot remove the intrinsic source-lag degeneracy.

The observation does not compare our domain with an inaccessible parent. Such a cross-domain comparison requires a transition map and dimensionless operational standard. The previous claim to have measured $c_{\text{parent}}/c_{\text{child}}$ from Ω_{Lambda} was circular and is retired.

DSF-2 now makes the permitted part of the question executable. Consider the covariant, disformal electromagnetic metric [43]

$$g_{\gamma,\mu\nu} = C(\Phi) g_{\mu\nu} + D(\Phi) \text{grad}_{\mu}(\Phi) \text{grad}_{\nu}(\Phi), \\ X = \text{dot}(\Phi)^2/2.$$

For a homogeneous field, relative to the gravitational/tensor metric,

$$c_{\gamma}/c_T = \sqrt{1 - 2 D X/C}, \\ \delta_T = c_T/c_{\gamma} - 1.$$

The conditions $C>0$ and $C-2DX>0$ preserve Lorentzian signature. If $D=0$, or if the field is stationary so that $X=0$, the cones coincide. A conformal change alone therefore cannot produce this signal. The implementation uses the dimensionless δ_T , never a raw numerical value of c , and evaluates the near-null expression without subtractive loss of precision.

The observational gate is deliberately literal, with fixed test vectors. A constant accessible late-time branch with $\delta_T=0$ is retained; branches with $\delta_T=+1e-14$ or $-1e-14$ are rejected by the quoted GW170817 interval. Those two nonzero values are test vectors, not predictions of CCT-1. CCT-1 has $X=0$ and returns the null control. This result excludes the named constant relative-cone subclasses along the relevant path; it does not test a scalar sound speed, a pre-geometric phase, or an inaccessible parent-domain speed.

8.5 Domain-setting and dark-sector tests

DSF-1 earns empirical content only by selecting a relation among independent dimensionless observables. Laboratory clocks already constrain present-day fine-structure drift. The Al+/Hg+ comparison found

$$\text{dot}(\alpha)/\alpha = (-1.6 \text{ plus_or_minus } 2.3)e-17 \text{ per year},$$

consistent with zero [46]. ESPRESSO absorption spectroscopy, combined with 28 measurements whose major wavelength-calibration errors were mitigated, found a weighted mean

$$\Delta \alpha/\alpha = -0.5 \text{ plus_or_minus } 0.5_{\text{stat}} \text{ plus_or_minus } 0.4_{\text{sys}} \text{ ppm}$$

over redshifts $0.6-2.4$ [47]. These are constraints under their stated measurement models, not a theorem that every inaccessible phase shares the same setting.

A future rolling one-field branch could become sharply testable. For example, if its independently derived electromagnetic coupling gives

$$\ln[\alpha(\Phi)/\alpha_0] = \zeta (\Phi - \Phi_0)/M_{\text{Pl}},$$

then a canonical homogeneous scalar obeys

$$[d \ln(\alpha)/d \ln(a)]^2 \\ = 3 \zeta^2 \Omega_{\Phi}(a) [1+w_{\Phi}(a)].$$

This would correlate clock/quasar spectroscopy with dark-energy evolution. It is a prediction only if the transition dynamics fixes ζ , the potential, and the initial state before the α and expansion datasets are used. The stationary CCT-1 limit has $\dot{\Phi}=0$, $w=-1$, no drift, and a common local cone. A nonzero late-time claim must improve on that null limit while obeying the clock, spectroscopy, CMB, equivalence-principle, and multimessenger bounds.

DST-1 faces a different test. Its $w=0$ background is necessary but not enough for dark matter. The same parameters must predict the matter transfer function, lensing potentials, halo scales, and abundance. Its $w=-1$ offset is necessary but not an explanation of dark energy until its value and stability are derived. A joint failure of either structure or expansion data rejects the corresponding physical identification while leaving the scalar identities intact.

DEB-1 tests the narrower statement that a literal boundary or outside contribution has a specified scaling. Normalize the alleged boundary length inside the child domain,

$$\text{ell}(a) = L_B(a)/L_B(a_{\text{ref}}) = [a/a_{\text{ref}}]^s,$$

$$\rho_B(a)/\rho_B(a_{\text{ref}}) = \text{ell}(a)^{-q}.$$

If this component is separately conserved, its continuity equation gives

$$\rho_B \text{ proportional to } a^{-q s},$$

$$w_B = -1 + q s/3,$$

$$\text{accelerating contribution in GR iff } q s < 2.$$

This turns the outside-force language into alternatives that can disagree with reality:

Literal scaling	q, s	Derived w_B	Result
Constant effective density	0, 1	-1	Accelerates; $\Delta \chi^2=1.2300$ from the BAO-only free- w minimum; background-degenerate with Λ
Constant-tension surface energy divided by volume	1, 1	-2/3	Accelerates; $\Delta \chi^2=11.3325$ in the BAO-only profile
Inverse-area density	2, 1	-1/3	Does not accelerate; $\Delta \chi^2=69.8429$ in the BAO-only profile

The surface row follows because fixed tension gives total energy proportional to area, E proportional to L_B^2 , while volume is proportional to L_B^3 , so ρ_B proportional to L_B^{-1} . The inverse-area row is a different ansatz, not another name for surface tension.

For a transparent observational pressure test, the repository records the published flat- w CDM marginal summaries from DESI DR2 [25]:

DESI+CMB	$w = -1.055 \pm 0.036$
+ Pantheon+	$w = -0.995 \pm 0.023$
+ Union3	$w = -0.997 \pm 0.027$
+ DESY5	$w = -0.971 \pm 0.021$

The version-0.5 summary gate compares each fixed w_B with the corresponding mean plus or minus 1.96σ . The $w=-1$ control lies inside all four approximate intervals; the $w=-2/3$ surface model lies outside all four, as does $w=-1/3$. These are one-dimensional published-posterior summary gates—not a reconstructed DESI likelihood, a model-independent measurement of pressure, or evidence that the surviving $w=-1$ term came from outside. DESI's time-varying- w model comparisons remain a separate, dataset- and parametrization-dependent issue.

The version-0.6 artifact is stronger and narrower: it evaluates the public BAO data vector and covariance rather than treating those four combined-data summaries as a likelihood. At fixed $w=-1, -2/3, -1/3$, the profiled chi-squares are 10.2710, 20.3735, and 78.8839, compared with the free minimum 9.0410. Raw $\Delta \chi^2$ is reported without converting it into a discovery claim. BAO alone constrains $H_0 \cdot r_d$, not H_0 , and a background likelihood cannot distinguish an external boundary from any other covariant mechanism with the same $H(z)$.

Finally, if energy crosses the boundary so that Q is nonzero, the separately-conserved identity above is inapplicable. That is a legitimate next model, not an automatic rescue: it must derive Q , its stress tensor, junction conditions, perturbations, and energy accounting before the target data are used.

9. What the existing computations do—and do not—show

The repository preserves every earlier script, but an execution and methods audit invalidated the former “thirteen reproducible tests” claim.

Former route	Live result	Scientific disposition
Compactness and entropy “consistency audit”	Mostly definitions plus hard-coded verdicts	Useful identity notebook; not empirical validation
DESI substitution	Recomputes hard-coded algebra under several published inputs	Sensitivity analysis only
de Sitter mode benchmark	Returns a value close to $n_s=1$ for a simple ODE	Narrow integrator regression, not certification of other equations
KS radiation background	Crashes from tuple indexing	Broken prototype
KS matter background	Overflows and produces non-finite states	Invalid numerical result
KS spectra	Depends on <code>/tmp</code> , enters invalid domains, lacks convergence	Invalid numerical result
Stiff-bounce spectrum	Returns about 3.005 , not advertised 2.94 , with ad hoc singular patches	Exploratory only
Junction tilt	Fits a toy coefficient and inverts observed n_s	Calibration, not zero-parameter prediction
CMB quadrupole	Wrong pixel permutation and invalid estimator/null procedure	Result and derived parent spin invalidated
Fertility scan	No local maximum, $p \approx 0.64$, then index crash	Negative result

Version 0.11.0 keeps the small standard-library package, controlled model laboratories/requirements layers, two data-facing artifacts, one restricted spherical matching audit, one homogeneous Einstein–Cartan finite-cap reduction, two GMF-1B closure exclusions, one spherical ECD axial-source preflight, one convention/action/interaction identity gate, the corrected algebraic torsion/contact-stress gate, and one bounded NGS-0 mathematical demonstrator. It verifies or calculates only:

- the Hubble-sphere compactness identity;
- the spherical H-SAT algebra;
- the distinction between inferred and independently predicted quantities;
- a transparent flat- Λ CDM age calculation;
- the exact CCT-1 closed-de Sitter background identities, finite invariants, and null-expansion classification;
- the IM-1 finite code-subspace isometry, complementary erasure reductions, and entropy bookkeeping;
- a deterministic classical scalar/tensor transfer matrix across a stated finite CCT-1 core, including Wronskian and refinement checks;
- the DST-1 scalar vacuum/oscillation effective-fluid identities, a compensated fixed- H spectator endpoint, and a matched scalar/radiation continuity ledger;
- the Kottler weak-field attraction/positive- Λ balance radius and sign change, explicitly not a horizon wall or recursive-domain result;
- GMF-1’s Oppenheimer–Snyder singular control and the same- Λ ,

finite- M Kottler-to-pure-CCT-1 direct Darmois obstruction, explicitly not a global child solution or shell-stability result;

- EC-1's reduced finite turning point, local trapped-to-normal-to-anti-trapped

interior sequence, and pressure-work obstruction to a fixed static vacuum boundary, not a finite-mass global attachment or an Einstein--Cartan--Dirac evolution;

- the restricted canonical Einstein--scalar null-convergence/Raychaudhuri

exclusion and the T^4-T^6 averaged-perfect-fluid sound-speed exclusion, not a full EC--Dirac characteristic analysis or a $1+1$ global evolution;

- the GMF-1B-ECD-SYM1 distinction between an axial-cancelling full-Dirac

singlet control and a Ventrella--Choptuik left-chiral spherical pair with a nonzero minimal-ECD contact invariant when both amplitudes are populated; the nonzero null torsion source but zero scalar invariant in either one-amplitude control; the unchanged metric-null Dirac principal cone; and a finite smooth annular initial source with exact spinor-zero buffers, not full-source equivariance, a complete effective stress, constraint-compatible vacuum region, bounce, or spacetime evolution;

- the GMF-1B-ECD-INT1 canonical-to-VC signature bridge, full contact-action

sign translation, real-current and nonlinear field-equation map, sphere-reduced action/direct-cubic weak identity, and tested-angle local retained-basis reconstruction, not a complete harmonic closure, the complete kinetic/spin-connection source, constraints, defocusing, bounce, or global solution;

- the GMF-1B-ECD-TOR1/KIN1 spin tensor, torsion invariant, full reduced-action

bookkeeping, 16-direction contact tetrad cancellation, zero contact null contraction, and one positive-frequency free-Dirac plane-wave control. The earlier fixed-current NEC result is retracted; no complete torsion-free Dirac stress, general Dirac NEC theorem, or bounce is claimed;

- NGS-0's finite spatial-band stationary point, general two-level recurrence

eigenvalue, and stable finite independent-trial probability. Its golden and non-golden fixtures derive no physical coefficients, cross-domain spectrum, dark sector, inevitable observer, or metaphysical conclusion;

- DSF-1 as an operational requirements framework for dimensionless setting and relative-cone claims, not as a computed variation;

- DSF-2's disformal relative-cone identity and conditional GW170817 gate, including passing and deliberately rejected fixed test vectors;

- DSF-3 as an H/Q transition-map and causal-calibration specification, not as

a computed cross-domain cone, varying local c , or infinite-speed result;

- DEB-1's separately conserved boundary-scaling identity and non-acceleration of an inverse-area density;

- the real 13-point full-covariance DESI DR2 BAO-only constant- w profile and fixed-scaling chi-squares;

- the exact-hash compact GW170817/GRB 170817A source-input validation, 1.737774 s timing reconstruction, and assumption-dependent published speed-bound arithmetic; and

- data-manifest sizes, hashes, semantic markers, and FITS header signatures for tracked/available inputs.

The new calculations do **not** verify a black-hole embedding, a microphysical phase transition, an exact gravitational factorization, a cross-domain spectrum, a dark-sector identity or origin, a cosmological golden ratio, inevitable observers, a metaphysical plan, a variable locally measured constant, a primordial curvature spectrum, or a new signal in data. Doing less is a gain when each output says exactly what it knows.

10. The minimal Recursive Horizons hypothesis

The active proposal can now be stated without excess machinery.

RH-0: Finite Gradient Closure

Every physical domain and transition has finite state data, finite local observables, and a declared conservation and entropy ledger. Under extreme localization, a derived invariant threshold activates a healthy restoring sector that grows faster than the focusing channel and produces a finite transition before geodesic incompleteness. Each domain has a lower closure and an upper closure or finite handoff. The recursive sequence may be unbounded in continuation without requiring any realized domain to be a completed infinite object. Nested domains are therefore gradients within gradients, not a license to extract perpetual work from equilibrium. A viable mechanism must derive how usable gradients are reorganized while constraints, quasi-local flux, and the appropriate generalized entropy law are obeyed. This is an H/Q programme constraint, not a theorem excluding mathematical infinity.

RH-1: Effective causal phase

There exists a dynamical state variable Φ_i such that a stable regime admits an effective Lorentzian metric and local invariant causal speed. In a pre-geometric regime, c is undefined rather than physically infinite.

DSF-1 makes this operational: the state must determine dimensionless couplings, mass and thermal ratios, or relative characteristic cones. A raw change in a dimensional number is not a prediction, and the recovered observable domain must satisfy the existing common-cone and constant-coupling limits.

DSF-3 adds the cross-domain burden. A finite handoff must map proper times, observers, phases, sectors, and states before it can compare a dimensionless relative cone or frequency calibration. A different lapse is not new physics; an unmapped dimensional c is not an observable; and a pre-geometric or degenerate limit is not a finite Lorentzian domain with infinite speed.

RH-2: FGC restoring transition

Under extreme collapse, the effective equations derive the FGC activation threshold and restoring sign, then admit a finite transition from a trapped interior branch to an expanding branch. The solution recovers classical GR away from the transition, survives anisotropy/shear tests, and derives an upper closure or finite handoff rather than stopping after a lower bounce.

CCT-1 now realizes the finite closed cosmological core and its null-expansion sign changes. GMF-1 establishes that direct same- Λ finite-M Kottler-to-pure-CCT-1 matching fails, while the Oppenheimer–Snyder control is singular. The Kottler benchmark proves that competing attraction and vacuum acceleration can define a scale in GR. None realizes the words “under extreme collapse”: a healthy restoring action, local Raychaudhuri magnitude, resolved exterior/interior junction, flux balance, upper closure, global causal structure, and invariant transition dynamics remain open. Einstein–Cartan–Dirac is one candidate audit, not RH-0 itself.

RH-3: Information-preserving map

A controlled quantum map carries state information through the transition without duplication. Its exact factorization and entropy accounting must be derived.

IM-1 demonstrates one nonduplicating finite isometry and exact entropy ledger under an assumed code-subspace factorization. The factorization and map have not been derived from gravity or CCT-1, so RH-3 remains a hypothesis.

RH-4: Effective child cosmology

The expanding branch approaches the successful hot-Big-Bang/ Λ CDM description over the observable domain, including local Lorentz invariance, early photons, BBN, the CMB, and structure formation.

DST-1 supplies a possible vocabulary for three pieces of that history—a vacuum-like offset, matter-like rapid oscillations, and interaction-driven radiation transfer. It supplies none of their observed abundances or perturbations. RH-4 requires those quantities to emerge from one connected solution rather than be attached after the fact.

RH-5: Optional horizon saturation

In the strong version, the child’s accessible asymptotic de Sitter entropy saturates the parent black-hole entropy budget, conditionally giving $\Lambda = 3/r_s^2$.

The first four propositions define the recursive-horizon research programme. RH-5 defines a particular thermodynamic branch. A physical dark-sector identification, baryogenesis, supersymmetry, a universal spin cascade, and inherited setting mutations are not needed to state the core. DSF-1 and DST-1 instead specify how the constants/dark-sector questions can be investigated without smuggling their answers into the premises.

11. How to prove the proposal wrong

A meaningful model must risk more than its metaphors. The following outcomes would kill or sharply restrict active branches.

11.1 Mathematical failure

- No ghost-free, stable effective action can produce the required causal phase.
- No invariant activation law produces a faster-growing restoring response with the required physical sign before singularity or EFT breakdown.
- Every candidate transition is singular, ill posed, or a coordinate artifact.
- Shear generically overwhelms the bounce mechanism.
- Every construction closes only the lower endpoint and leaves the expanding domain without an upper closure, finite handoff, or finite-state operational completion.
- No consistent global extension connects the exterior, trapped branch, and expanding branch; in particular, the direct same- Λ finite- M Kottler-to-pure-CCT-1 no-shell attachment fails.
- No quasi-local flux and generalized-entropy ledger closes without creating energy or extracting perpetual work from equilibrium.
- The quantum map duplicates information or cannot define its physical sectors.
- Every domain-setting effect reduces to a unit change, coordinate artifact, or conformal rescaling with no relative observable.
- Every proposed cross-domain cone comparison lacks a physical clock/phase/sector transition map, or reduces to a bare dimensional c_i/c_j .
- The proposed coarse-domain limit loses hyperbolicity, stability, Lorentzian signature, EFT control, or no-signalling rather than producing a finite relative characteristic.
- No global flux/junction solution turns the local balance intuition into a trapped-to-expanding geometry.

Any of these would reject the corresponding bridge even though black-hole thermodynamics and GR remain intact.

11.2 Cosmological failure

- The derived perturbations cannot reproduce the observed adiabatic nearly scale-invariant scalar spectrum.
- The model spoils BBN, acoustic peaks, lensing, structure growth, or local Lorentz invariance.
- A rigid H-SAT branch predicts constant dark energy and decisive, robust data establish incompatible evolution.
- A pre-registered unique anisotropy or waveform template fails on appropriately blinded data.
- The scalar excitation has the wrong abundance, lifetime, sound/Jeans scale, lensing, or structure growth, or its transfer sector fails reheating, BBN, and CMB constraints.
- A claimed setting variation violates clock, spectroscopy, equivalence-principle, CMB, or multimessenger limits.

11.3 Predictive failure

- Every proposed observable is fixed by the same data it claims to predict.

- The theory reduces exactly to standard Λ CDM and classical black-hole GR for all possible observations while adding no operational content.

In the latter case the framework may remain an interpretation, but it is not a distinct physical model.

12. A concrete calculation programme

12.1 Background

1. Choose spherical symmetry only as a first controlled problem.
2. Work in horizon-regular variables.
3. Specify matter, spin/torsion or quantum corrections, and an equation of state.
4. Integrate with adaptive events and constraint monitoring.
5. Demonstrate convergence under tolerance, grid, gauge, and boundary variation.
6. Calculate curvature invariants and both null expansions.
7. Extend to anisotropy and rotation before connecting to astrophysical black holes.
8. Reproduce GMF-1 first: calculate induced/extrinsic (or null transverse) data, the Misner–Sharp jump, and both expansion pairs for a declared interface. Then derive a resolved action/layer, horizon-crossing flux, and parent/child junction balance; do not substitute the local Kottler radius or a prescribed shell for this global result.
9. Evolve the DST-1 scalar and its interaction products self-consistently through reheating, radiation, matter, and late acceleration.
10. For DSF-3, derive every relevant principal symbol and the transition map

for proper time, observer frame, phase, field sector, and state; report a mapped dimensionless relative cone or mark the domains uncomparated.

12.2 Perturbations

CCT-1 permits a limited but honest linear calculation. In conformal time η ,

$$\begin{aligned} a(\eta) &= L \sec(\eta), \\ a'/a &= 2 \sec^2(\eta) - 1, \\ -\pi/2 < \eta < \pi/2. \end{aligned}$$

Because the background field is constant, $\delta \Phi$ is gauge invariant at first order and decouples from scalar metric perturbations. For scalar harmonics on the unit three-sphere,

$$-\nabla^2 Q_{nlm} = n(n+2) Q_{nlm}, \quad n=0,1,2,\dots$$

the canonical mode $u_n = a \delta \Phi_n$ obeys

$$u_n'' + [(n+1)^2 + (m_\Phi L)^2 - 2] \sec^2(\eta) u_n = 0.$$

For transverse-traceless tensor harmonics in the stated convention,

$$-\nabla^2 G_{ij}^{(n)} = (n^2 - 3) G_{ij}^{(n)}, \quad n=3,4,5,\dots,$$

the canonical tensor amplitude obeys

$$\mu_n'' + [n^2 - 2 \sec^2(\eta)] \mu_n = 0.$$

The closed- S^3 eigenvalues and tensor equation follow the convention derived explicitly by Bonga, Gupta, and Yokomizo [40]; their scalar index is shifted by one relative to the $n \geq 0$ label used here. The positive canonical scalar

kinetic term, positive Einstein tensor kinetic term, and unit principal propagation speeds give the no-ghost/no-gradient-instability result for this baseline [32,40]. They do **not** establish bounded evolution of every infrared mode: for sufficiently light, low- n scalar modes the displayed effective frequency squared can be negative over part of the core. That is a tachyonic/infrared growth question rather than a ghost or gradient instability, and any bounded-mode claim must specify the mass, harmonic, finite interval, and amplitude criterion.

The repository integrates both independent basis solutions across a declared interval $[-\eta_{\text{core}}, +\eta_{\text{core}}]$. Since neither equation contains a first-derivative term, the resulting real 2×2 fundamental matrix must have determinant one. Step refinement and an exactly soluble constant-frequency scalar case test the implementation.

This is a **classical symplectic transfer**, not yet a quantum Bogoliubov coefficient or a primordial spectrum. The endpoints are not asymptotically Minkowskian, no preferred adiabatic in-vacuum has been supplied, and the background with $\text{dot}(\Phi) = 0$ is not a single-clock source for the observed comoving-curvature perturbation. It therefore predicts neither n_s nor A_s . The next perturbation step is to:

1. specify an evolving, invariant transition background and a regular gauge-invariant scalar variable;
2. derive rather than insert its scalar, vector, and tensor quadratic actions;
3. define an initial quantum state only where its adiabatic interpretation is valid;
4. derive reheating and conversion into the adiabatic curvature mode;
5. publish converged machine-readable spectra; and
6. propagate the fixed result through a Boltzmann code and likelihood.

12.3 Information

IM-1 completes the finite linear-algebra checks. The physical work remains:

1. define relational observable algebras, gravitational dressing, boundary/edge data, and a quantitative code error;
2. show that constraints permit the claimed approximate sector factorization;
3. derive the channel from covariant transition dynamics and its backreaction rather than choosing p ;
4. retain the distinction between fine-grained, generalized, and horizon thermodynamic entropy; and
5. derive whether any saturation condition follows.

12.4 Observation

For an **additional strict globally closed-child branch postulate**, the observable child retains positive spatial curvature $K=+1$ after the controlled core. In the usual cosmological convention this supplies the exact conditional sign statement

$$\Omega_{K=+1}(t) = -c^2/[a(t)^2 H(t)^2] < 0 \quad \text{when } H \text{ is nonzero.}$$

This is the first independent **conditional** discriminator: its sign is fixed by the added global branch postulate, independent of H-SAT, and was not inferred from a target curvature measurement. CCT-1 alone does not derive the matching from its closed de Sitter core to the observable child topology. A robust $\Omega_{K=+1} > 0$ inference that survives a stated family of nonflat cosmological models and systematic checks would falsify the strict branch. An observational result statistically consistent with zero, or a small inferred negative value, would not confirm it; subsequent inflation or expansion can drive the magnitude arbitrarily close to zero, and many unrelated closed cosmologies predict the same sign. Planck plus BAO is consistent with near spatial flatness [19,21].

The sign test is therefore independent but weak. A stronger version must fix the magnitude through a complete post-bounce expansion history. Closed spatial sections also make the perturbation spectrum discrete; a completed initial-state and reheating calculation could predict correlated low-mode structure. Those parameters must be frozen before looking at the target data. If CMB anisotropy is later chosen, the analysis must specify release, maps, masks, multipole range, covariance, signed amplitude, axis convention, and null optimization before unblinding; every searched choice must recur in end-to-end null simulations.

DSF-1/DST-1 add two pre-registration targets. First, an action may fix a correlated $\alpha(a)-w(a)$ curve rather than an arbitrary drift; clock and spectroscopy data then test the curve while expansion data test its paired equation of state. Second, a genuinely nonconformal matter/gravity metric may fix a relative photon/tensor cone; multimessenger propagation then tests its sign, scale, and frequency dependence. If all such functions are fitted to the same target observations, neither is independent. If the complete model fixes them upstream and they survive the recovery tests, they are stronger than the current curvature sign because they connect otherwise separate experiments.

DSF-3 adds a later, stricter target. If one completed transition identifies the relevant clocks, phases, sectors, and observers on both sides, it may fix the mapped ratio

$$R_{ab}^{(i \rightarrow j)} = [(c_a/c_b)_j] / [(c_a/c_b)_i].$$

That prediction must be set upstream and propagated to an accessible observable. A lapse ratio, coordinate speed, bare c_i/c_j , conformal compactification, or entanglement correlation cannot substitute for it.

Version 0.7 evaluates the data-facing rejection machinery on exact public inputs and adds the GMF-1 direct-match obstruction before the transition derivation is complete. DSF-2's source record reconstructs the GW170817/GRB 170817A delay and the assumption-dependent interval that rejects constant accessible relative-cone values outside it. DEB-1's real DESI DR2 BAO-only profile gives $\Delta \chi^2 = 11.3325$ for the constant-tension surface case and 69.8429 for the nonaccelerating inverse-area case, while $w = -1$ is only 1.2300 above the free constant- w minimum. These are meaningful negative results for literal subclasses. They are not an independently fixed prediction of the global theory: no transition action currently fixes D_X/C , q , s , or a nonzero exchange Q . The next observation-grade step is to derive those values upstream and then apply a prospectively frozen analysis without retuning them.

13. The philosophical layer

The historical project describes reality as energy, information, God, code, simulation, yin and yang, or the Force. These images express a monist intuition: there is one reality, and our mathematical languages are ways of compressing its stable structure. They may inspire scientific questions, but they are not interchangeable scientific claims.

Physics does not currently identify energy as a programmer capable of overriding its own equations. "Simulation" can mean at least three different things:

1. the universe is mathematically describable;
2. effective laws emerge from deeper dynamics;
3. an external agent literally computes our world.

The first is ordinary scientific practice, the second is a research hypothesis, and the third is metaphysical unless it yields an operational discriminator. A religious interpretation may coexist personally with the physical programme, but it cannot be used as evidence in the derivation.

The project's strongest personal reading can nevertheless be stated without mockery: energy—or the reality whose conserved changes we describe as energy—may be understood as God, creator, and the source of every form that can arise. On that reading, the nested process is a plan and our capacity to study it is part of the process itself. Physics neither confirms nor refutes that meaning. It asks a different question: do the proposed transition, spectral, flux, and complexity relations follow from equations and survive observation? Keeping those questions distinct lets spiritual meaning motivate the work without making it a premise that protects the model from failure.

The "eternal opportunity" intuition also has a precise conditional form. If each sufficiently independent finite domain has a fixed nonzero probability p of producing a declared life-like complexity class, the probability of at least one occurrence after N trials is $1 - (1-p)^N$, which approaches one as N grows without bound. Eternity alone does not prove $p > 0$, independence, ergodicity, or even a probabilistic ensemble, so almost-sure occurrence is not logical necessity. NGS makes deriving that measure and recurrence part of the scientific task.

The phrase "math is built into reality" can be given a modest scientific reading: reproducible relations exist, and mathematics captures them. Whether mathematics is invented, discovered, or ontologically prior is not decided by the horizon relation.

The project's preferred interpretation of infinity is similarly philosophical but coherent: infinity need not be a completed spatial object; it may describe an unending process or a limit. Physics still allows models with infinite spatial slices. Recursive Horizons need not deny those models to explore causal genealogies.

14. Why Einstein and Hawking could rest easy together

The subtitle is not a declaration of victory over physics. It names a design constraint.

Einstein's lesson is that gravity is causal geometry, locally expressed through a Lorentzian metric and tested with extraordinary precision. Hawking and Bekenstein's lesson is that horizons are also thermodynamic and informational objects. The project does not need local light to slow, energy to evaporate into nothingness, or black holes to violate thermodynamics. It asks whether the classical endpoint is a phase boundary whose deeper completion respects both sides.

Version 0.7 adds a third lesson already implicit in both: geometry responds to stress-energy, and the same covariant equations can contain opposed tendencies. Positive vacuum energy and compact mass can define a balance scale; a scalar can present vacuum-like, matter-like, and transfer behavior in different regimes. This is how "it can be both" becomes physics: not by treating two words as interchangeable, but by deriving them as limits of one state and one set of equations.

It also adds a discipline that matters just as much: two elegant pictures can produce different scalings. A constant effective density, a surface energy, and an inverse-area density are not interchangeable versions of "outside pressure." Their equations of state and acceleration properties differ, and the repository now records where the literal alternatives pass or fail.

If the answer is yes, Einstein's geometry remains the effective description away from the transition, Hawking's thermodynamics constrains the information budget, and quantum dynamics supplies the missing bridge. If the answer is no, the attempt still clarifies why the ingredients were insufficient.

That is a productive reconciliation: nobody is asked to be wrong in the domain where the evidence says they are right.

15. Conclusion

Recursive Horizons now has a stronger scientific thesis than a collection of analogies: Finite Gradient Closure. A causal domain may be a stable phase in which geometry, dimensionless settings, vacuum stress, matter excitations, transport, and information settle into one reproducible relationship. A black-hole interior and an expanding cosmology may be opposite causal branches of that same relationship. This is not a denial of established physics. It is a proposed global completion of theories whose local success is a mandatory boundary condition. FGC is still H/Q: its toy scale selectors, regular-core controls, and candidate matter sectors organize the work but do not derive the law. NGS is the preferred constructive language for what finite closure could generate: QFT modes and spectral density, EFT resolution and coarse-graining, thermodynamic occupation and dissipation, and a finite inherited complexity budget. It does not yet identify the dark sector, guarantee observers, or derive a universal hertz or golden-ratio spacing.

The phrase "we see things for what they truly are" now names the project's method, not a declaration that its preferred ontology is already true. Established observations retain their force while the ontology used to explain them remains contestable. Every gradient, boundary, relational, or dark-sector variable must earn explanatory status through an invariant map, baseline recovery, a prospectively changed observable, and a falsifier. Antimatter is kept with the directly evidenced sectors; stable asymmetry remains a derived stability and finite-throughput question. That is how the programme stays open without becoming unfalsifiable.

Version 0.11.0 moves another layer below analogy. NGS-0 makes finite-band selection, a general two-level recurrence eigenvalue, and the conditional finite-trial probability executable without treating their fixtures as cosmology. DSF-1 says exactly what a domain setting must mean operationally. DSF-2 demonstrates a covariant route to a relative cone; exact-hash source records reconstruct the published GW170817 timing arithmetic and its conditional speed interval. DSF-3 now states what the deeper finite-domain idea would have to add: an action-derived principal structure and a

mapped dimensionless cone/clock transfer. It rejects the shortcuts that expansion itself changes local c , that a coarse level has completed infinite signal speed, or that CCC, QED vacuum effects, and entanglement already demonstrate such a level. DEB-1 rejects an inverse-area term as a source of acceleration. The real DESI DR2 BAO-only profile strongly disfavors that fixed $w=-1/3$ background and the constant-tension surface-only $w=-2/3$ term, while retaining $w=-1$ as a background-compatible but origin-degenerate control. DST-1 proves the canonical scalar's vacuum-like, matter-like, and transfer identities under declared assumptions. The Kottler benchmark proves that attraction and negative-pressure acceleration can meet at a finite but unstable sign-change radius. CCT-1 gives a regular covariant closed core and derived null-expansion regions. IM-1 gives an exact nonduplicating code-subspace isometry. The perturbation module gives a converged classical transfer matrix. The strict closed-child branch commits to the conditional sign $\Omega_K < 0$. GMF-1B-PF1 then removes two tempting shortcuts before they are mistaken for a transition: restricted canonical Einstein--scalar matter cannot null-defocus, and the homogeneous thermal EC perfect-fluid closure does not remain a causal/hyperbolic radial material. GMF-1B-ECD-SYM1 then identifies one action-level candidate that survives those two filters: a classical left-chiral two-spinor candidate with an $SO(3)$ -compatible axial-current sector, a nonzero minimal-ECD axial contact invariant when both amplitudes are populated, an unchanged metric-null Dirac principal cone, and clean compact initial spinor support. It does not prove full ECD source equivariance or yet derive the complete stress, regular-centre constraints, or a spacetime. GMF-1B-ECD-INT1 now removes a convention ambiguity before that source is placed in a solver. It explicitly translates the canonical $(+---)$ minimal-ECD action and Hehl--Datta equation to the Ventrella--Choptuik $(-+++)$ representation and verifies that the reduced contact-action variation matches the directly projected cubic interaction. TOR1/KIN1 then verify the algebraic torsion/contact sector and retract a false fixed-current defocusing result. The physical contact stress has zero null contraction. This closes the contact subproblem, not the torsion-free Dirac stress, Einstein constraints, or evolution. The constructive target is a finite cap, while a complete S^3 child or any inner-horizon passage remains conditional on further global and stability work.

Together these results show that the picture is mathematically composable and that some literal implementations already fail. They do not show that nature composes it this way. The joining relation remains the new physics: a global horizon-regular collapse solution, an invariant domain-balance transition, derived setting/coupling functions, a physical information channel, reheating and abundance, realistic scalar/tensor perturbations, and one independently fixed observable correlation. The scalar background does not yet identify dark matter or dark energy; a surviving constant-density background does not identify an external source; the Kottler radius does not create a parent boundary; H-SAT remains a conditional clue; current CMB, DESI, clock, spectroscopy, multimessenger, and code results do not confirm the genealogy.

The next move is therefore direct and ambitious, but ordered. FGC-1-ACT1 has frozen one smallest candidate family and its linear activation invariant; FGC-1-VAR1 has now expanded and cross-checked its covariant bulk metric source. FGC-1-HYP1-RED1 has now evaluated those equations in a general radial ADM spherical chart and exposed the complete uneliminated second-jet structure. FGC-1-HYP1-SYM1 has now quotiented the local covariant radial symbol in two overlapping Bianchi-row patches, derived the pointwise matter and regulator characteristic factors, constructed exact scalar companion symmetrizers, and rejected one defective fixed-gauge shortcut. FGC-1-HYP1-MHG1 has now frozen a direct modified-harmonic principal formulation, separated the unphysical pure-gauge and gauge-constraint cones from the physical factors, and proved complete real pointwise eigenbases for the full frozen radial principal matrices. Its formulation boundary is compared with the relevant Horndeski hyperbolicity, well-posed modified-harmonic, and ESGB evolution literature [79, 80, 81]. FGC-1-HYP1-MHG2-REF1 has now frozen a tensorial flat spherical reference gauge on a positive-radius annulus, evaluated the complete gauge extension, and recovered MHG1's principal gauge block from its exact second-jet derivative. FGC-1-HYP1-MHG3-IMP1 has now proved one exact local coordinate-time acceleration branch through a flat inactive FGC-QR vacuum/reference-gauge root, with the nonzero action couplings retained. FGC-1-HYP1-MHG4-PROP1 has now derived and cross-checked the exact conditional lower-order reference-gauge operator on independent spherical C^μ , including complete coefficient extraction and MHG1 principal-column regression. FGC-1-HYP1-FO1-RC1 has now supplied the exact local eighteen-variable first-order implicit lift, all complete-residual argument derivatives, the complete flat-root branch linearization, MHG1 principal regression, and the chosen kinematic radial reduction identity. FGC-1-HYP1-DOM1-QIFT1 then gives a continuous exact rational enclosure at fixed $r=4$: all thirty local mixed/spatial two-jet parameters vary independently within half-width $1/65536$, every parameter point has a unique acceleration root in the radius- $1/128$ box, the contraction bound is below $1/250$, and the minimum image-box inclusion margin exceeds $3/512$. FGC-1-HYP1-CON1-COMP1 then constructs one exact activated local two-jet inside that box. Two selected

spatial jets solve $H_{\text{ww}}=0$ and $\text{nabla}_r C^r=0$; frozen choices and exact coefficient identities give $M=C^\mu=\text{nabla}_r C^t=0$. Composing these facts with QIFT1's unique acceleration root and the nonsingular normal gauge-extension map forces the remaining gauge derivatives and obtains the undefined metric and both scalar equations at that point. The post-COMP1 records now prove the complete radial point-mode construction, a compact nonflat uniformly invertible radial eigenframe and symmetrizer, the direct metric-derived gauge subsidiary identity, conditional boundary-free gauge preservation, and frozen radial main-system maximal dissipation. The exact DEF0 observable also prevents auxiliary gauge cones or a negative R_{kk} term alone from being called physical affine defocusing. SRC1-NL1 additionally verifies binary64 evaluation and analytic acceleration differentiation through the same complete REF1 tensor code and a safeguarded QIFT1-local branch solve. CON4-PHY1 then derives the undefined Hamiltonian/momentum constraints and their acceleration-free structure, proves the exact invertible REF1-shell map to normal gauge data, composes CON2/CON3 into conditional boundary-free physical/gauge/reduction propagation, and defines executable normalized monitors. These are substantial local/radial advances. CTR1-REG1 additionally supplies the exact rescaled regular-centre REF1 Taylor formulation with finite curvature and explicit defect controls. ID1-FAM1 then solves the complete specialized FGC-QR physical constraints directly on a maximal polar--areal slice, verifies exact full-evaluator and gauge controls, and constructs a regular-centre finite-mass nonzero open local family with independent χ data and an explicit nonzero ϕ seed. These results still do not supply a nonlinear constraint-preserving quasilinear IBVP or a spacetime trajectory. DOM4-RUN1 now adds a nonzero classical parameter/state container, the complete initial-slice bridge, and strict same-equation acceleration witnesses. HYP2-MD1 adds the complete 12-field covariant principal symbol and a coefficient-complete sufficient principal-health ball for every spatial covector, with exact spherical-symbol agreement. It is explicitly not retained-EFT authorization, nonlinear existence, trajectory containment, or nonspherical evolution/robustness. EFT1-OPEN1 makes the physical-promotion stop quantitative: only three of twelve necessary retained-EFT/PDE run predicates pass. RUN1-SYM1 now makes the narrower mechanism-test boundary equally explicit. Its outcome-neutral FGC-2-SF1 protocol passed its structural schema audit, but ID0-PREF1 proves that PROTO1 is not executable as an initial-data study: it omitted Π_ϕ , and its whole original amplitude box lies below the declared compactness floor under the minimal completion. The frozen file remains as the failed premise record. **FGC-2-SF1-PROTO2** performs the first premise-only, pre-holdout repair, freezing $\Pi_\phi=0$, the GR-0-only amplitude list, and a new output namespace; the validator requires every other PROTO1 rule unchanged and ID0 binds its chronology to immutable checkpoint `d4f0cc8...`. ID1 then exposes the independent nine-eighths-width/centre-buffer contradiction. Preserved **FGC-2-SF1-PROTO3** verifies the immutable PROTO2/ID1 blobs at `fae2c5b...`, changes only the floor to the exact $39/16$ L_0 positive vacuum, versions the future namespace, and requires an ordered static all-case premise ledger before GR-0 dynamic eligibility. SRC1 and CON4 close only two of eight complete scoped predicates at the immutable pre-ID1 checkpoint. Active RUN1 also consumes CTR1 and ID1, closes the combined centre/data predicate, then consumes DOM4 and HYP2 to close the combined classical run-domain/multidirectional early-kill predicate. BND2-CP1 closes the scoped boundary row through a strict all-cone causal budget and HLT1-MON1 closes the transactional classical-health row on direct and injected controls. NUM1-VAL1 then closes the independent numerical and affine- measurement row with two method families and outcome-blind analytic controls. This yields the preserved $7/8$ structural record, but CAL0-PREF2 now proves that PRO3-HLD1 cannot be honestly resolved: the declared compact pulse fails PROTO3's literal last-bin admission at every grid despite rapidly convergent top-band power, and target-specific health limits lack a GR-0/FGC-QR applicability map. No candidate evolution outcome was opened. **FGC-2-SF1-PROTO4** and **FGC-1-PRO4-FRZ1** now freeze the premise-only successor by immutable PROTO3/CAL0 lineage. They preserve the action, inputs, observable, outcome rules, and robustness box while partitioning all 26 legacy stops and replacing only the defective spectral/constraint admission with convergence and conservative error budgets. This is test design, not a dynamical result. **FGC-1-HLT2-MON2** implements the branch-owned stop map, proper weighted spectral tests, common-event three-grid constraint admission, and additive-only numerical-error accounting. Its static controls pass, but it opens no trajectory. **FGC-1-ID2-ALL1** then completes the pre-calibration Cartesian ledger across all seven amplitudes, five future FGC-QR modifiers, both methods, and three grids. Its 252 hash-bound complete states give 31/35 FGC-QR static passes and freeze $[5/2, 3]$ as the only candidates for fresh calibration under a repaired protocol. The four FGC-QR and three GR-0 exclusions are static contract findings, while the disclosed 17/35 nested FGC-QR diagnostic does not rewrite PROTO3's five-Boolean manifest. **FGC-1-CAL1-PREF3** then performs the missing native semidiscrete $t=0$ composition for both eligible amplitudes, both methods, and all three grids. It proves that literal PROTO4 cannot admit its declared experiment for test-contract reasons, demonstrates the minimal outcome-blind repair without reading any trajectory, and requires a separately frozen PROTO5. PRO5-FRZ1 supplies the immutable freeze, and HLT3-MON3 binds the exact GR-0 runtime, twelve fresh inputs, four passing $t=0$ common events, and the recoverable campaign transaction.

CAL2-PREF4 reduces the resulting first campaign: both amplitudes stop before the first nonzero common event on the same threshold-adjacent source gate, while checkpoint replay proves the accepted state and first half-step retry pass the unchanged raw gate. This is a PROTO5 trial-stage stop-ownership obstruction, not a GR-0 or FGC-QR result. PROTO6/PRO6-FRZ1 freeze only that numerical ownership repair, and HLT4-MON4 implements it. CAL3-PREF5 now binds the completed recalibration and proves that the accepted state passes while only a wholly unaccepted k_4 /endpoint proposal misses; one further halving passes. PROTO7/PRO7-FRZ1 freeze the general transaction, HLT5-MON5 implements it, and CAL4-PREF6 now binds the completed campaign. Both amplitudes reach the first nonzero common event; CAL4 then localizes the shared veto to projector/coarse-witness ownership for amplitude $5/2$, while amplitude 3 remains independently inadmissible. PROTO8/PRO8-FRZ1 now freeze that ownership correction under unchanged thresholds. HLT6-MON6 now implements the correction without changing the inherited trajectory transaction and authorized one fresh GR-0 calibration. CAL5-PREF7 now binds the completed campaign: amplitude $5/2$ stops at $t=1/8$, amplitude 3 at $t=1/16$, and neither is eligible. The repaired ownership succeeds on evolved data, but fixed constraint gates and the amplitude-3 derivative tail remain terminal. CAL5 therefore motivates only a separately frozen one-level PROTO9 resolution ladder under unchanged physics and thresholds; its conditional 8193 estimates are not results. PROTO9/PRO9-FRZ1 freeze that exact full-ladder change. HLT7 then constructs all twelve direct inputs: sources, constraints, and absolute spectra pass, but all four new finest-pair nested-tail admissions fail at $t=0$. The runtime binding exists while fresh calibration remains closed. CAL6-PREF8 now performs the outcome-blind diagnostic-sensitivity adjudication: every raw failure persists, but it lies below a strictly contracting map-conditioning scale and the complete profiles converge. This is not a continuum-error bound or a calibration pass. PROTO10/PRO10-FRZ1 now freeze that premise-only interpretation without opening a trajectory. Its HLT8 runtime certificate and the resolved holdout manifest remain open at that freeze boundary. HLT8-MON8 now supplies the separate runtime certificate: it exactly preserves all four raw failures, admits all four guarded $t=0$ controls, and fails closed under every removed prerequisite. It authorizes one fresh GR-0 calibration only. CAL7-PREF9 now binds its completed record and shows that neither amplitude reaches an evolved common event: the fine member exhausts the unchanged source-only retry path on a centre-adjacent binary64 ϵ_{64}/r^2 floor. Its successful exact-reference calculation is a $t=0$ control, not runtime evidence. PROTO11/PRO11-FRZ1 freeze the exact-reference map, and HLT9 now implements and attacks it: all twelve raw source checks and four complete $t=0$ ladders pass. CAL8-PREF10 binds the resulting campaign: both amplitudes evolve to $t=1/16$, but $5/2$ later meets an exact affine source/minimum-step wall and 3 independently fails the binding direct coarse ϕ/Λ derivative-tail veto. No amplitude is eligible. SRC2-CAP1 subsequently reproduces the complete implicated replay and captures its affine fixture; none of the frozen CAP1 routes clears the gate. SRC2-PREF11 then proves at the first positive-radius node that the exact dyadic source system is nonsingular and that both floating residual paths falsely reject the correctly rounded exact root. This localizes that specific wall to numerical cancellation while changing no claim about trapping, collapse, FGC-QR, a Raychaudhuri margin, or the physical mechanism. FGC-1-SRC3 now supplies the reference-covariant evaluator, matches exact reference and nontrivial oracle controls, and clears the original source gate across the captured full grid without changing the equations or threshold. No new trajectory is contained in that certificate. FGC-1-RSP1-FRZ1 freezes six fresh amplitude-3 states on 4097 $\rightarrow$ 8193 $\rightarrow$ 16385 and authorizes one outcome-neutral run. FGC-1-RSP1-PREF12 now binds and independently recomputes the completed endpoint: all four direct target ratios pass $<1/4$, all six members reach $t=1/16$ with zero source retries, and the targeted constraint, profile, budget, and causal premises pass. Only the amplitude-3 veto is cleared for successor design. The generic all-field raw spectral aggregate remains false, calibration remains incomplete, no case is eligible, and PROTO12 had not yet been frozen at the RSP1 boundary. FGC-1-SRC4-VEC1 now adds a prospectively frozen tensor-contracted backend for the unchanged SRC3 source map. Exact-reference, exact-dyadic, independent-backend, adversarial, and CAP1 controls pass while every equation, threshold, reference choice, root branch, and claim remains fixed. Its local speedup is engineering context only, and it opens no trajectory. FGC-1-PRO12-FRZ1 now binds SRC4 and the immutable RSP1 checkpoint into the raw-preserving pairwise diagnostic contract. Both methods qualify that contract without converting their false generic raw admissions into direct convergence. FGC-1-HLT10-MON10 now implements and attacks that exact runtime, reconstructs all twelve inputs and four complete time-zero events, and authorizes one fresh outcome-neutral PROTO12 GR-0 calibration. FGC-1-CAL9-PREF13 now binds the completed 49-event execution. Both amplitudes reach late evolved common events with zero serialized source retries, but neither is eligible because only the terminal SSPRK3 radial-momentum order falls below the unchanged $3/2$ gate. Permitted adaptive CFL reductions are tracked separately and did not cause either terminal. The matching vetoes do not prove a common cause, and no SGB-L/FGC-QR outcome, DEF1 result, or physical claim was read. FGC-1-RSP2-FRZ1 supplied the required prospective freeze and runtime authorization: only the new amplitude-3 SSPRK3 16385 member could

advance, while the CAL9 4097 and 8193 states remained immutable inputs and the $t=23/16$, raw-constraint, roundoff-enclosure, and $p \geq 3/2$ contracts remained unchanged. FGC-1-RSP2-PREF14 now binds the completed run and independently reconstructs target orders 1.4990947384363291 and 1.9817436463819333, with complete constraint admission passing. This permits only a separate prospective PROTO13 freeze. FGC-1-PRO13-FRZ1 now supplies that freeze: amplitude 3, six complete $t=23/16$ restart payloads, method-owned resolution ladders, unchanged final time 32, and unchanged scientific gates. HLT11 implements and authorizes that exact restart runtime. FGC-1-CAL10-PREF15 now binds its completed terminal at event 63: both method-owned spatial/constraint admissions remain true with zero source retries, while both methods fail both layers of the first 64-sample temporal-spectrum gate. The trapped sign remains false. The checkpoint independently reproduces the stop but does not identify whether its cause is arithmetic, normalization, finite-window/taper behavior, sampling, interpolation, a resolved transient, or continuum regularity. The terminal checkpoint cannot resume; only a separately frozen temporal-gate diagnosis may follow, and no GR-0 case is eligible. FGC-1-TDG1-FRZ1 now supplies that prospective freeze. Before reading the terminal histories it proves by synthetic counterexample that normalized raw temporal tail fractions erase amplitude convergence and therefore cannot be a general spatial-convergence observable for nonzero continuum histories. The $1/4$ PROTO13 result remains historical evidence; only a bounded read-only diagnosis is authorized, with no replacement gate or trajectory. FGC-1-TDG1-PREF16 now binds that diagnosis. It confirms the normalized-ratio defect on all six actual ladders, while 395 above-floor individual failures, 49 below-order unnormalized comparisons, and 134 interpolation-limited comparisons leave the absolute-tail question open. Only a prospectively frozen TDG2 discriminator may follow; no replacement gate, PROTO14, eligible GR-0 case, candidate, or physical claim follows. FGC-1-TDG2-FRZ1 supplies that prospective freeze without reading the actual histories. FGC-1-TDG2-PREF17 now binds the mixed outcome: some finest tails contract toward zero, a small subset is conditionally nonzero, and 101 field plus 102 derivative failures remain enclosure dominated. The binary64 DFT audit owns none of them; the conditional interpolation debit dominates the unresolved majority. FGC-1-TDG3-FRZ1 prospectively freezes two native no-resampling estimators and passes every synthetic discriminator control. FGC-1-TDG3-PREF18 now binds their actual-history result: the robust zero-contracting subset survives, the tentative nonzero subset does not, and common-grid interpolation is not the sole unresolved owner. No continuum enclosure is supplied. FGC-1-TDG4-FRZ1 now prospectively freezes the exact sampling-kernel theorem, independent smooth and uniform-alias witnesses, and the quantitative assumptions required to control the unobserved between-sample directions. Its no-history controls pass. FGC-1-TDG4-PREF19 now completes the theorem, retires the old rule only as continuum admission, retains it as a sampled diagnostic, and leaves CAL10 unreclassified. It authorizes only prospective replacement-gate design. FGC-1-TDG5-FRZ1 now freezes one finite-dimensional complete-state Hermite and same-grid step-refinement architecture under that permission. Its exact no-history controls pass. FGC-1-TDG5-PREF20 independently completes the exact finite-dimensional map/restriction/extremum theorem and confirms that the immutable engine exposes a fresh endpoint right-hand side. FGC-1-TDG5-IMP1 now implements and synthetically validates both atomic method-owned pairs and their continuous outward binary64 candidate/Bernstein envelope. FGC-1-TDG6-FRZ1 now freezes the exact three-level $p \geq 3/2$ contraction rule and complete additive finest-pair debit without an absolute or physical-signal-scaled threshold. FGC-1-TDG6-PREF21 now independently binds the exact quarter-map, interval-order, channel/debit, and immutable runtime-feasibility results and authorizes implementation plus synthetic qualification only. FGC-1-TDG6-IMP2 now implements and synthetically qualifies the resulting seven-path compositor, including independent tracer shadows, complete channel admission, fine-only atomic debit commit, durable temporal retries, and checkpoint restoration. FGC-1-PRO14-FRZ1 now binds the precise successor protocol to immutable commit 27badf9... and all six exact $t=23/16$ restart payloads. The retired 64-sample continuum admission remains diagnostic only; every accepted new step instead requires the complete TDG6 test. The debit begins at restart and supplies no pre-restart temporal or physical error bound. FGC-1-HLT12-MON12 now implements and attacks the temporal-aware production integration. It restores all six restart payloads, binds the pinned runtime plus CAL11 plan and runner, verifies fine-only commit, separate retry ownership, rollback, and checkpoint recovery, and creates neither future namespace. CAL11-PREF22 now binds the first terminal attempt: the event-24 RK4-2049 proposal failed the TDG6 binary64 subdivision validator before a shadow or accepted macro step existed, leaving the atomic checkpoint at event 23 with all restart hashes and zero temporal ledgers unchanged. The calibration is invalid rather than completed. FGC-1-TDG7-FRZ1 now freezes the prospective stage-safe exact-binary64 shared lattice: $G=8Q$ makes actual RK4/SSPRK3 $c=\{0, 1/2, 1\}$ stage times and endpoints exact, unlike the historical rounded boundaries. It derives the arithmetic root cause without repair, PROTO14 mutation or resume, or fresh calibration. FGC-1-TDG7-PREF23 now independently binds the theorem and authorizes only later runtime-repair implementation plus synthetic qualification. TDG7-IMP3 now implements and synthetically qualifies the repair without history access; the repair opens no

protocol/namespace, calibration, eligibility, candidate, or physics. Its stateless retry limitation permits only prospective replacement-protocol freeze design with a durable cursor. FGC-1-PRO15-FRZ1 now freezes that lifecycle without implementing it: sealed IMP3 and six restart payloads bind a campaign-owned durable `FRESH_READY/RETRY_PENDING` cursor. Nonexhausted `TDG6TemporalRetryRequired` records rejection then exact successor before retry; exhausted `TDG6TemporalRetryExhausted` records a terminal invalid/nonconverged checkpoint and never creates `RETRY_PENDING` or a physical classification. Post-rejection recovery reconstructs the exact typed branch or stops invalid. Canonical compact-JSON/float-hex hashes bind complete TDG6 ledger and journal tip; checkpoint publication requires file, replace, and parent-directory `fsync`; rollback creates a new generation from the restored logical snapshot instead of rewinding old cursor bytes. It reads, mutates, and resumes no PROTO14 raw history/checkpoint, creates no namespace, and authorizes/runs no fresh GR-0 calibration. Eligibility, SGB-L, FGC-QR, DEF1, EFT, and physical claims remain false. FGC-1-HLT13-MON13 now completes only the campaign-owned durable runtime and deterministic synthetic fault qualification required by that freeze. It uses content-addressed states, a framed hash-chained journal, and complete six-member generational checkpoints to enforce byte-exact executed-plan acceptance, exact retry/exhaustion closure, terminal lock, deterministic recovery, and rollback through new generations rather than reused cursor bytes. Its executed $110=6+14+90$ nominal/mutation/fault corpus also replays both old and new directory images at ambiguous replace windows. It creates no production runner, run plan, namespace, trajectory, calibration, eligibility, SGB-L, FGC-QR, DEF1, retained-EFT, transition, or physical result. `PROTO15_pretrajectory_runtime_authorized=false` and `PROTO15_fresh_GR0_dynamic_calibration_authorized=false` remain binding; PROTO14/CAL11 are unresumed invalid-runtime history. A separately authorized production runner, externally trusted genesis, run plan, and prelaunch authorization are next. Local hash integrity does not authenticate wholesale replacement of a self-consistent store. FGC-1-PRO16-FRZ1 now freezes the missing successful common-event transition and its external trusted-genesis requirements. A `COMMON_EVENT_COMMIT` may occur only after all six `FRESH_READY` members reach the active target with zero current TDG6 retries. Its receipt is strictly predecessor-only: the outer record hash exists before successor cursors or checkpoint hashes are derived, removing the self-hash cycle. Future genesis evidence must expose six explicit descriptors; their aggregate hash is only a checksum. **FGC-1-PRO16-PREF24 statically diagnoses that complete successor cursor/high-water copy rules and a complete external generation-zero construction are not yet frozen.** At that historical PREF24 boundary this was not HLT14, a production adapter or runner, namespace, trajectory, calibration, GR-0 eligibility, SGB-L, FGC-QR, DEF1, retained-EFT, transition, or a physical result. HLT14 was then unauthorized and only a prospective PROTO17 freeze could follow. FGC-1-PRO17-FRZ1 freezes that missing construction as a pure reference map: the full six-member genesis ledger/state/cursor/high-water/checkpoint mapping and the acyclic predecessor-only common-event successor were fixed before the later HLT14 implementation. It creates no namespace and authorizes neither a trajectory, calibration, candidate result, nor physical conclusion. FGC-1-PRO17-PREF25 independently reconstructs that complete abstract synthetic mapping without importing the PROTO17 implementation and matches its generation-zero, receipt, and successor hashes. FGC-1-HLT14-MON14 now completes the production-shaped implementation qualification only on temporary synthetic stores, with $72=4+6+18+44$ unique cases. It separately reproduces the sealed hashes, authenticates temporary authority, and fully rehashes temporary real-NPZ inputs. Raw hashes are input-dependent, and its exception-fault coverage is not a power-loss proof. Actual raw-byte validation, launch-time Git/blob/live-import authority, historical payload replay, and production namespace reuse/foreign-store rejection remain `required=true, completed=false`. Only separate production-prelaunch design was authorized at that historical HLT14 boundary; no namespace, pretrajectory, calibration, candidate, EFT, or physical conclusion followed. FGC-1-PRO18-PREF26 subsequently completed raw-byte/hash recomputation and semantic historical-payload replay for the declared actual sources. FGC-1-PRO18-AUTH1 constructs canonical commit-independent, calibration-only authority content. At that historical boundary it was not self-authenticating and advanced no state. FGC-1-HLT15-GEN1 has now used immutable AUTH1 commit `2e3373a...` to validate the external Git/blob/live-byte tuple, numerical environment, imports, and actual six-state reimport before first output-root observation and atomic generation-zero installation. All six members remain `FRESH_READY`, with zero accepted macro-steps, retries, or debits. FGC-1-PRO18-PREF27 independently authenticates that exact eight-leaf store and proves reuse, foreign-store, malformed-tree, staging-residue, and arriving-root no-clobber behavior on disposable same-filesystem roots. All four HLT14 production duties are complete, but state advance, calibration, eligibility, candidate, EFT, transition, and physical claims remain closed. Only a separately frozen first bounded progression contract may be designed. FGC-1-PRO19-FRZ1 / FGC-2-SF1-PROTO18 freezes that edge, and FGC-1-HLT16-MON16 qualifies durable payload, descriptor, journal, complete-checkpoint, retry/recovery, status, bounded-runner, and committed one-event preflight machinery. The first

authenticated attempt completed five finite accepted **RK4-2049** macro-steps and stopped invalid before a CFL-retry record was published because attempted-cap identity was serialized incorrectly. Clean nonterminal generation 6, checkpoint **6277b3be...**, is the exact recovery boundary. FGC-1-PRO19-CFL1-FRZ1 freezes one prospective corrected-image continuation subject to post-commit image/store authentication. No common event, GR-0 calibration, candidate execution, activation, trapping, DEF1, retained-EFT promotion, transition, or physical claim exists. FGC-1-PRO19-SID1-FRZ1 now authenticates the actual generation-7 pre-recovery suffix: its complete checkpoint remains at sequence 6, while sequence 7 is a TDG6 rejection with no accepted state. It freezes the sole sequence-8 cursor/ledger transition and complete generation-8 successor. That metadata-only recovery has now been published after immutable SID1 commit **c83ef958...**. FGC-1-PRO19-SID2-PREF1 independently binds its parent chain, clean namespace, temporal retry cursor, unchanged six descriptors, and bitwise-preserved affected **u, p, q** arrays. FGC-1-PRO19-SID3-AUTH1 now binds the exact committed GR-0 execution closure owned by immutable implementation commit **4d7133c...** and the exact compact generation-8 inputs. Shared analytic dependencies may contain dormant candidate-capable definitions and default parameter records, while no candidate-only module or evaluator, runtime configuration, initial or evolution state, branch, or outcome is opened. SID3 does not inspect the live mutable store. The guarded preflight later passed and one bounded resume appended only generation 9's generic **HLT16AttemptError** invalid-terminal record and terminal lock. FGC-1-PRO19-PREF28 binds that exact edge and proves the accepted member map is unchanged from generation 8; the persisted record does not expose its exception message or root cause. No common event, calibration, SGB-L, FGC-QR, activation, trapping, DEF1, EFT, transition, mechanism, or physical claim is earned. The resolved holdout manifest remains open. SGB-L now has branch-owned source/control instruments but no complete source-admission, IBVP, or aggregate-health certificate. The conversion from normalized constraint residual to a DEF1 observable error now has candidate-blind conversion/error-map instruments, but no separate FRZ1/PREF1 or trajectory-dependent bound. A nonlinear constraint-preserving IBVP and full EFT1 promotion remain separate stronger burdens. FGC-2-SF1 may then consume a valid COL1 trajectory and return either a finite, resolution-independent positive complete Raychaudhuri margin or a converged, robustly scoped obstruction. A crash, unconverged solve, or failed isolated point is not a scientific negative. A branch rejection must name the exact parameter-box x data-family x formulation scope, converge in an invariant or canonical dimensionless quantity, survive both methods and gauge/domain/boundary perturbations, and have analytic or independent-solver support. Finite held-out cases alone establish no open neighborhood; ROB1 must provide an interval or continuity certificate over the frozen rational physical box. DEF0-OBS1 is only the exact pointwise measurement contract. The independent ECD-ID1 branch remains frozen during this campaign rather than competing for interpretation. FGC-3 must solve the global flux/junction problem, upper closure or handoff, entropy ledger, invariant continuation, and any inner-horizon stability gate. RUN1 hard-locks physical-transition authorization false even if local DEF1/ROB1 succeeds. NGS-1 through NGS-4 must then derive the proper-frequency mode transfer, finite capacity, coarse-grained stress, any DSF-3 causal calibration, and a frozen dimensionless discriminator; any golden-ratio eigenvalue must emerge from that transfer rather than be selected. Only then should the model evolve the scalar and its products through the hot history, derive structure and the adiabatic spectrum, and freeze a dimensionless cross-dataset prediction before looking at its target data. If a candidate construction fails, that bridge is removed while the organizing question and established ingredients remain; if no viable action can pass the FGC gates, the law itself fails as physics. If it succeeds, Einstein's geometry, Hawking's horizon thermodynamics, quantum information, and cosmology will no longer sit beside one another as separate clues; they will be limits of one tested process. That is the claim worth pursuing—and now it has equations sharp enough to disagree with reality.

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